

Compressing Global Dynamics of Deep Nonlinear Feature Learning

Amir Joudaki

September 27, 2026

Abstract

How much can the training dynamics of a deep network be compressed? Compressing a trained function does not by itself compress its learning: networks that agree on every input can drift apart once training resumes. We ask how small a state suffices to reproduce the *whole* training trajectory while keeping depth, effective nonlinearity and feature learning, because the state that survives is a candidate for the variables that govern learning. History-based descriptions such as Tensor Programs and dynamical mean-field theory remove explicit width dependence while retaining temporal response objects. We show that the history itself is compressible. Measured in a clock that runs at the speed of learning, each neuron’s forward and backward response history is summarized by q shifted-Legendre moments. The resulting autonomous system has a fixed number of coordinates throughout a prescribed finite horizon. At every fixed width and finite depth, under the stated activation regularity, it tracks the whole parameter trajectory with error $O_T(q^{-1})$, or $O_T(q^{-2})$ with a second, joint clock. Pairing forward and backward memories makes the compression defect a product of two projection errors. As a corollary, every learned hidden-weight increment on that horizon lies within $O_T(q^{-1})$ of a rank- mq matrix generated online from response moments, where m is the sample count. The initialized matrices are retained. Small orders closely reproduce dense fitted predictions in whole-circle RMS and on held-out MNIST inputs, and outperform the frozen and directly trained low-rank alternatives tested on the circle tasks. Placed on a common axis of evolving state against accuracy, existing descriptions of training pay for the limit in neurons, in stored history or in tuples of samples; response memory pays in an approximation order.

1 Introduction

A missing model of learning. The Turing machine is the canonical model of computation. It is idealized—it has no fixed bound on memory—and it is simple enough to reason about, while physical computers, from the von Neumann architecture onward, are understood as its finite-resource implementations. One might view neural networks as playing the role in learning that physical computers play in computation; what this analogy still lacks is a counterpart to the Turing machine—a canonical idealized model of learning. Deep neural networks trained by gradient descent are the obvious candidates for implementations of such a model, but the object they would implement is not yet known. The obstacle is that their training dynamics are enormously complex and have no simple representation.

The limit, and why to compress it. A natural place to look for the idealized object is the limit of infinite width. Under feature-learning scalings of the type used in tensor-program and mean-field theory (Yang and Hu, 2021; Nguyen and Pham, 2023), width-dependent learning rates preserve order-one feature evolution, and suitable population observables obey deterministic limiting laws. The limit is the network with no resource constraint, as the Turing machine is the computer with

none; a finite network would then be this limit plus a quantified deviation from it, as in a central limit theorem. Going to infinite width to simplify seems paradoxical, since the dynamical system grows with width. Statistical physics resolves the paradox in other settings: a gas of 10^{23} molecules is simpler to describe than a gas of ten, because its macroscopic behaviour is captured by a few collective variables. Training presents two sources of complexity: the number of neurons, and the response history introduced when evolving weights are eliminated from the description. The weights themselves are a Markovian state; response-based limits express their learned interactions through the past. Mean-field and tensor-program limits address width. Whether the response history can also be *compressed* is our question. We do not seek compression for its own sake: a compressed description must retain what drives learning and may discard the rest, so the state that survives is a candidate for the variables that govern learning, and a compressed system that reproduces training would be a step toward the idealized model.

What the model must keep. Several influential simplifications isolate different aspects of training. Deep linear networks admit exact solutions but remove the nonlinearity (Saxe et al., 2014); one-hidden-layer mean-field models remove multilayer hidden interactions (Chizat and Bach, 2018; Mei et al., 2018; Sirignano and Spiliopoulos, 2019); and the neural tangent kernel keeps depth and nonlinearity but freezes the features (Jacot et al., 2018; Lee et al., 2019; Chizat et al., 2019). Each is simple precisely because of what it gives up. Our target keeps all three: *depth* (more than one hidden layer), *nonlinearity* that stays effective during training and in the limit, and *feature learning* (hidden representations move at leading order). It must also reproduce the whole trajectory, not a snapshot: two networks that compute the same function on every input can have different Jacobians and backward responses, and once training resumes the discrepancy accumulates. Together with a specified population limit, this contract fixes the object we seek to understand.

The gap, and one axis for every description. Width compression leaves a further resource question. Tensor Programs carry variables indexed by every earlier training step (Yang and Hu, 2021), and feature-learning dynamical mean-field theory (DMFT) evolves two-time kernels whose memory grows quadratically with the number of time points (Bordelon and Pehlevan, 2022). These descriptions encode learned interactions through temporal responses rather than explicit learned matrices, a substantial reduction in width dependence, but their state grows with the number of retained time points. We ask whether evolving learned interactions can instead be carried by a fixed number of neuron-level coordinates, with a bound on the error of the complete training trajectory. The question suggests one axis on which every description of training can be placed: fix a horizon and an accuracy, and ask how much *evolving* state a description needs to reproduce the trajectory to that accuracy, and whether that state grows with the number of elapsed steps. Descriptions that freeze the features, drop depth or drop the nonlinearity target a different training regime, and enlarging their state alone supplies no general approximation guarantee for ours. Those that keep the contract each pay for the limit in a different currency: the dense network in neurons, Tensor Programs and DMFT in stored history, the Neural Tangent Hierarchy in tensors indexed by tuples of training samples. Response memory pays in an approximation order: one order serves a chosen horizon, and no coordinates are added merely because more steps elapse; its required value can depend on that horizon and on the task. Definition 2.1 makes the axis precise, Table 1 (Section 5) places the descriptions on it once the results are in hand, and Appendix E works out, under stated hypotheses, what each would cost at a prescribed accuracy.

Key contribution. Compressing the training dynamics of a deep, fully connected network means carrying its trajectory with less evolving state than its dense weight matrices. Written without those matrices, gradient flow is exact but its state is the entire response history, and exact finite closures of it run into a concrete obstruction (Section 3.1). We show that this history can nevertheless be compressed *in time*: for a network of any fixed width and depth, replacing each neuron’s forward and backward history by its first q Legendre moments in a clock that runs at the speed of learning yields an autonomous system whose state is fixed on the chosen horizon and whose trajectory stays within $O_T(q^{-1})$ of gradient flow, or $O_T(q^{-2})$ with a joint clock (Theorems 3.1 and 3.2). The difficulty is feedback—the memorized signals are produced by the approximation itself—and controlling it along the whole trajectory is the content of the proofs. As a consequence, every learned weight increment lies within $O_T(q^{-1})$ of a rank- mq matrix generated online from the network’s own response moments (Corollary 3.3), and in experiments small orders reproduce the fitted function where frozen kernels and rank-matched trained factors do not (Section 4). The theorems are finite-width statements; the population version, which would be the idealized model this introduction asks for, is stated as Conjecture 3.4.

2 Setting

Fix m training pairs (x_a, y_a) with $x_a \in \mathbb{R}^d$, width n and $L \geq 2$ hidden layers:

$$\begin{aligned} z_a^{(1)} &= W^{(1)} x_a / \sqrt{d}, & h_a^{(1)} &= \phi(z_a^{(1)}), \\ z_a^{(\ell)} &= W^{(\ell)} h_a^{(\ell-1)}, & h_a^{(\ell)} &= \phi(z_a^{(\ell)}), \quad 2 \leq \ell \leq L, \\ f_a &= \frac{w^\top h_a^{(L)}}{n}, & r_a &= f_a - y_a, \quad \rho = \left(\frac{1}{m} \sum_a r_a^2 \right)^{1/2}. \end{aligned} \tag{1}$$

The activation ϕ may differ between layers; the loss is $\mathcal{L} = \rho^2$. The backward responses, which exclude the residual, are $\delta_a^{(L)} = w \odot \phi'(z_a^{(L)})$ and $\delta_a^{(\ell)} = \phi'(z_a^{(\ell)}) \odot (W^{(\ell+1)})^\top \delta_a^{(\ell+1)}$. With block mobilities $(n, 1, \dots, 1, n)$, gradient flow reads

$$\dot{W}^{(1)} = -\frac{2}{m} \sum_a r_a \delta_a^{(1)} \frac{x_a^\top}{\sqrt{d}}, \quad \dot{W}^{(\ell)} = -\frac{2}{nm} \sum_a r_a \delta_a^{(\ell)} (h_a^{(\ell-1)})^\top \quad (2 \leq \ell \leq L), \quad \dot{w} = -\frac{2}{m} \sum_a r_a h_a^{(L)}. \tag{2}$$

We write $\theta = (W^{(1)}, \dots, W^{(L)}, w)$ for all parameters and $\dot{\theta} = F(\theta)$ for (2). Hidden entries are initialized as $\mathcal{N}(0, 1/n)$, first-layer entries as $\mathcal{N}(0, 1)$, and the stored readout is small.

Population flow. As $n \rightarrow \infty$, the neurons of layer ℓ become a probability space Ω_ℓ with expectation $\mathbb{E}_\ell$: hidden coordinates $z^{(\ell)}, h^{(\ell)}$ become random variables in $L^2(\Omega_\ell)$, the first-layer weights a random row $W^{(1)} \in L^2(\Omega_1; \mathbb{R}^d)$, the readout $w \in L^2(\Omega_L)$, and each hidden matrix a bounded operator $W^{(\ell)} : L^2(\Omega_{\ell-1}) \rightarrow L^2(\Omega_\ell)$ used together with its true adjoint. The same equations (2) hold with $\frac{1}{n} \sum_i$ replaced by $\mathbb{E}_\ell$ and outer products $\delta h^\top / n$ by the rank-one operators $g \mapsto \delta \mathbb{E}[h g]$. Such limits are established in several scopes—for fixed depth on a short horizon, and globally in special models—with finite-width gradient flow converging to them on compact horizons; a global theorem for general data and depth is open. Where we need the population flow we state its existence as part of the model.

What counts as compression. An exact compression reproduces the reference trajectory; an approximate one reproduces it up to a controllable error. A compression may also match the current

state and velocity while its small errors compound until the trajectories separate; we ask for the stronger, global notion.

Definition 2.1 (Global approximation on compact horizons). Autonomous systems indexed by an order q , with trajectories θ_q , *globally approximate* a reference flow θ if for every $T < \infty$

$$\sup_{0 \leq t \leq T} \|\theta_q(t) - \theta(t)\| \longrightarrow 0 \quad (q \rightarrow \infty).$$

The rate in q quantifies the trade-off between the size of the compressed state and its accuracy.

The size in question is the *moving state* of a description: the numbers it updates while computing training, excluding fixed sources such as the initialization or a frozen kernel.

3 Closing the training dynamics: exact and approximate

3.1 Exact closure: operators and an obstruction

A closed description of training evolves a state of one of three kinds: finitely many *scalars*, independent of width and training time; finitely many *populations* (functions, random variables or laws on fixed spaces, such as the law of a neuron’s weights), whose particle implementation stores one value per particle and field; or *operators* between population spaces, the limits of matrices, which record how each row relates to each column, a coupling that one-neuron marginal statistics discard. A finite number of named fields or operators need not contain finitely many scalars, so every claim below names both the type and the size of the retained state.

Before allowing errors we ask what exact capture of the population flow can look like. A natural candidate without growing history is an operator ODE: (2) read with the hidden matrices $W^{(\ell)}$ as operators. In a deep linear benchmark—one input, three linear hidden layers, order-one initialization—this can be proved completely: the learned part of the whole network becomes a single trace-class operator that evolves by an explicit autonomous cubic equation driven by the residual, and finite gradient flow and gradient descent converge to it on every compact horizon in the stated limits (Appendix D). But one named operator can contain infinitely many degrees of freedom, so an operator equation alone supplies no finite state–accuracy trade-off; such a trade-off requires a further approximation of the operator. For a two-hidden-layer nonlinear population flow on the circle one can likewise describe the dynamics exactly through *current* observations—joint laws of finitely many expressions built from the current weights, the initialization, ϕ , and applications of $W^{(2)}$ and its adjoint—but the derivative of each finite level of this hierarchy introduces higher-level observations, and the construction does not supply an exact closure at a fixed finite level (Appendix C).

At the other extreme, in the same linear benchmark no encoder formed from a fixed finite list of tensor contractions of the weights with width-independent bounded degree admits a polynomial autonomous scalar ODE and polynomial readout that reproduce the output dynamics on an open set of states at all large widths (Appendix D). The reason is that repeated differentiation of the output forces ever larger connected contractions: the $(2j - 1)$ st derivative contains $\frac{1}{n} (W^{(1)})^\top ((W^{(2)})^\top (W^{(3)})^\top W^{(3)} W^{(2)})^j W^{(1)}$, of degree $4j + 2$, which no bounded-degree alphabet can express. The result is restricted to this encoder and readout class; it identifies a concrete obstacle already present with linear features, not a no-go theorem for all finite scalar descriptions. Between the two, whether the exact operator can be compressed into finitely many populations is open: rank-one gradient updates do not by themselves imply a fixed rank bound on their time integral. The main result takes a constructive route: accepting controlled error gives an explicit autonomous system with a quantitative state–accuracy trade-off.

3.2 Approximate closure: the idea

Integrating (2), every hidden matrix is

$$W^{(\ell)}(t) = W_0^{(\ell)} - \frac{2}{nm} \sum_a \int_0^t r_a(s) \delta_a^{(\ell)}(s) (h_a^{(\ell-1)}(s))^\top ds. \quad (3)$$

The learned part is an integral of backward against forward responses. Two direct representations are the dense matrix and the accumulated response history. A familiar analogy is a moving ball: under a known force law, its current position and velocity can determine the future without storing its past positions. Here we seek history coordinates that retain enough information for the network’s own law of motion. Regular response histories suggest that a few coefficients can summarize the interaction integral. Two things must hold: the summary must be *closed*, so that it evolves autonomously, and *accurate* over the whole of training, which alternates between bursts of learning and plateaus.

3.3 Closure: shifted Legendre moments

Let p_k be the shifted Legendre polynomials on $[0, 1]$, normalized by $p_k(1) = 1$ and $\int_0^1 p_j p_k = \mathbf{1}_{j=k}/(2k+1)$. Their derivatives reduce to lower modes,

$$x p'_k(x) = k p_k(x) + \sum_{j < k} (2j+1) p_j(x), \quad (4)$$

and this is what makes the moment system close exactly. The identity underlies online polynomial memory in HiPPO (Gu et al., 2020); the new element is that the memorized signals are the network’s own forward and backward responses, coupled back through the reconstructed network. The first moments of a history are its mean, its linear trend (the first polynomial rises from -1 at the start to 1 now) and its quadratic shape; each added moment corrects what the previous ones miss (Figure 8, Appendix H).

3.4 Accuracy: a clock that follows learning

Concentrated variation can require many polynomial modes in physical time. We therefore measure history in a clock τ that runs at the speed of learning,

$$\dot{\tau} = \rho, \quad \tau(0) = 1 : \quad (5)$$

fast when the residual is large and slow as it approaches zero. This speed is the square root of loss, not its time derivative; it need not stop at a positive-loss stationary point. The unit prefix $[0, 1]$ carries the initial forward responses and no backward response. For each hidden link $\ell \geq 2$ and sample a we store the first q Legendre moments, in the clock coordinate, of the forward response $h_a^{(\ell-1)}$ and of the residual-weighted backward response $r_a \delta_a^{(\ell)}/\rho$:

$$\bar{h}_{a,k}^{(\ell-1)} = \int_0^\tau h_a^{(\ell-1)} p_k(\xi/\tau) d\xi, \quad \bar{\delta}_{a,k}^{(\ell)} = \int_0^\tau \frac{r_a \delta_a^{(\ell)}}{\rho} p_k(\xi/\tau) d\xi, \quad 0 \leq k < q,$$

where ξ is the clock value at which each response was recorded. By (4) these moments obey the closed system

$$\begin{aligned} \dot{\bar{h}}_{a,k}^{(\ell-1)} &= \rho h_a^{(\ell-1)} - \frac{\rho}{\tau} \left[k \bar{h}_{a,k}^{(\ell-1)} + \sum_{j < k} (2j+1) \bar{h}_{a,j}^{(\ell-1)} \right], \\ \dot{\bar{\delta}}_{a,k}^{(\ell)} &= r_a \delta_a^{(\ell)} - \frac{\rho}{\tau} \left[k \bar{\delta}_{a,k}^{(\ell)} + \sum_{j < k} (2j+1) \bar{\delta}_{a,j}^{(\ell)} \right], \end{aligned} \quad (6)$$

with $\bar{h}_{a,0}^{(\ell-1)}(0) = h_a^{(\ell-1)}(0)$ and all other moments zero, and the hidden matrices are reconstructed as

$$\widehat{W}^{(\ell)} = W_0^{(\ell)} - \frac{2}{nm\tau} \sum_{a=1}^m \sum_{k=0}^{q-1} (2k+1) \bar{\delta}_{a,k}^{(\ell)} (\bar{h}_{a,k}^{(\ell-1)})^\top. \quad (7)$$

All responses in (6) are evaluated in the current *reconstructed* network, and the first layer and readout follow (2). The system is autonomous and never divides by ρ . Equivalently, with Π_q the $L^2(0, \tau)$ projection onto polynomials of degree below q ,

$$\widehat{W}^{(\ell)} = W_0^{(\ell)} - \frac{2}{nm} \sum_a \int_0^\tau \left(\Pi_q \frac{r_a \delta_a^{(\ell)}}{\rho} \right) (\Pi_q h_a^{(\ell-1)})^\top d\xi, \quad (8)$$

and since cross terms vanish by orthogonality, the error in each matrix update is a *product* of a forward and a backward projection error. Figure 7(b) (Appendix H) illustrates this cancellation and distinguishes it from the separate stability argument that controls feedback.

The joint feature-learning clock. The learning-speed clock makes forward histories smooth but not every backward history. A second clock advances with the residual *plus* the speed of all forward and backward responses,

$$\dot{\tau} = \rho + \left\| \frac{d}{dt} (h_a^{(\ell-1)}, r_a \delta_a^{(\ell)} / \rho)_{\ell,a} \right\|_2, \quad \tau(0) = 1, \quad (9)$$

so that the whole response path moves at unit speed at most in the clock coordinate. History is placed by this clock but still weighted by the residual mass ρdt , so the Legendre modes are no longer orthogonal and one shared $q \times q$ Gram matrix is tracked alongside the moments; the reconstruction uses its inverse, the defect keeps the product form, and the clock speed can be evaluated explicitly by one directional forward and backward pass (Appendix B). The coordinate change is illustrated in Figure 4 (Appendix B).

3.5 Global guarantees

Theorem 3.1 (Learning-speed clock). *Fix finite width, depth $L \geq 2$, arbitrary finite data and a finite initialization, and let every activation be in $C_{\text{loc}}^{1,1}(\mathbb{R})$. For every $T < \infty$ and all sufficiently large q the closure exists uniquely on $[0, T]$ and*

$$\sup_{0 \leq t \leq T} \left\| \widehat{\theta}_q(t) - \theta(t) \right\| = O_T(q^{-1}),$$

where the threshold on q and the implied constant depend on T , the width, depth, data and initialization but not on q . The rate extends to hidden responses and to predictions on any bounded set of unseen inputs. If ϕ and ϕ' are bounded ($\tanh$, sigmoid, arctangent, sine), it holds for every $q \geq 1$. No assumption of successful fitting, Gaussian initialization, input geometry or label structure is made.

Theorem 3.2 (Joint feature-learning clock). *Under the setting of Theorem 3.1, assume the activations of layers $\ell \geq 2$ are in $C_{\text{loc}}^{2,1}$ and $\rho(0) > 0$. For every T and all sufficiently large q the closure exists uniquely on $[0, T]$, its clock stays bounded independently of q , and $\sup_{t \leq T} \left\| \widehat{\theta}_q(t) - \theta(t) \right\| = O_T(q^{-2})$.*

What separates this from approximating a given signal, as in HiPPO, is feedback: the histories being memorized are produced by the network reconstructed from the memories, so approximation errors change the signals they approximate. Both proofs (Appendices A and B) rest on three facts: an exact identity for the growth of the projection error of each history; the product structure of the defect; and the Legendre tail bound $\|u - \Pi_q u\|^2 \leq \tau^2 \|u'\|^2 / (4q(q+1))$. With the learning-speed clock the forward histories have q -independent derivative bounds while backward histories only need bounded mass, giving $O(q^{-1})$; with the joint clock the whole path is unit speed once the clock is shown to be bounded, both factors contribute, and the rate is $O(q^{-2})$. Two steps carry the difficulty. In the learning-speed clock, bounding forward histories through the layers requires the reconstructed velocity, whose memory endpoint term grows like q ; it is cancelled exactly by the q^{-1} projection factor, which is why the rate does not degrade with depth. In the joint clock, the clock itself must be bounded without assuming it, which we do through an order-independent bound on the total variation of the trajectory. Gronwall's inequality and a first-exit argument then transfer the accumulated defect to the trajectory.

Corollary 3.3 (Learned increments are low-rank in their own history). *Under Theorem 3.1, for every T and all sufficiently large q , each dense learned hidden increment $W^{(\ell)}(t) - W_0^{(\ell)}$, $2 \leq \ell \leq L$ and $0 \leq t \leq T$, is within $O_T(q^{-1})$ in Frobenius norm of the matrix $-\frac{2}{nm\tau} \sum_{a,k < q} (2k+1) \bar{\delta}_{a,k}^{(\ell)} (\bar{h}_{a,k}^{(\ell-1)})^\top$, which has rank at most mq and is built from the moments of the closure's own responses. Under Theorem 3.2, the joint-clock reconstruction, of rank at most $m(q+1)$, is within $O_T(q^{-2})$.*

The corollary follows from the theorems because the reconstruction (7) has rank at most mq . An a posteriori singular value decomposition gives the best approximation at a chosen rank but does not itself bound that rank or produce a closed training dynamics. Here the approximation is *generated autonomously and online* from the network's own forward and backward history, with a rank that does not grow with time: on a given horizon, one fixed order serves every time inside it, however much the features have moved. The statement is informative only while $mq < n$, and its constant depends on width.

3.6 Cost and size at accuracy

With L hidden layers, width n , m samples and order q , each learned $n \times n$ increment is carried by $2mq$ vectors of length n . The moving internal state is $2(L-1)mnq$ numbers against $(L-1)n^2$ for the dense learned matrices; it does not grow with training time and is smaller whenever $2mq < n$. The initialized matrices, $(L-1)n^2$ numbers, are kept as a fixed source. Per step, applying $W_0^{(\ell)}$ and its transpose costs $O(mn^2)$ per layer as in dense training, the learned correction $O(m^2nq)$ per layer, and the moment update $O(Lmnq)$; the joint clock adds an $O(q^3)$ Gram factorization and $O(Lmnq^2)$ solves.

Read as a trade-off, Theorems 3.1 and 3.2 say that, at fixed width and horizon, a trajectory error ε costs $O(mn/\varepsilon)$ moving numbers per hidden layer with the learning-speed clock and $O(mn/\sqrt{\varepsilon} + 1/\varepsilon)$ with the joint clock. Two comparisons on the same network are exact. The learned increment can always be carried by its own history, since summing the updates $-\frac{2}{nm} r_a \delta_a^{(\ell)} (h_a^{(\ell-1)})^\top$ over the steps taken is an identity; that costs mn numbers per step and grows with training. And the dense increment itself is n^2 numbers. Against the first, the order q replaces the number of steps; against the second, $2mq$ replaces n . Both are consequences of the theorems, not of a hypothesis. What they do not say is how the constants grow with the horizon or the width, and this is why the population-level version of the comparison (Appendix E) is conditional.

What is compressed is time, not samples or neurons. The state grows linearly with the number of training samples, and once $2mq \geq n$ it exceeds the dense learned matrices; learning from a

continuum of inputs would need a further representation of the input dependence. Nor is this a runtime improvement: because $W_0^{(\ell)}$ still acts densely, the closure ran somewhat slower than dense training in our experiments.

3.7 From finite networks to the population

The theorems hold at every fixed width, for every realized initialization. At the population level they suggest the following picture, which we state as an interpretation, not a result. For each layer and sample, the q forward and q backward moments are random variables over neurons, and the learned action of $W^{(\ell)}$ becomes $g \mapsto -\frac{2}{m\tau} \sum_{a,k} (2k+1) \bar{\delta}_{a,k}^{(\ell)} \mathbb{E}_{\ell-1}[\bar{h}_{a,k}^{(\ell-1)} g]$: finitely many neuron-level memory fields, coupled through the initialized operator. A width- n network would then be a Monte Carlo simulation of this system. What is proved is only a diagonal statement: if finite-width flows converge to a population flow on $[0, T]$, choosing $n_j \rightarrow \infty$ and then q_j large enough at each width yields systems in the response-memory family converging to it. This diagonal argument alone does not ensure $mq_j < n_j$ or a compression gain. The natural next theorem is the following.

Conjecture 3.4 (Fixed-order population closure). *For each fixed q , as $n \rightarrow \infty$ the order- q response-memory closure converges on compact horizons to a limiting system consisting of the initialized operators and finitely many neuron-level memory fields per layer and sample; and this limit tracks population gradient flow with error $O_T(q^{-1})$, or $O_T(q^{-2})$ with the joint clock, uniformly in width.*

The limit in the conjecture would be a time-local, finite-order description of deep nonlinear feature learning in the population limit: the kind of object the introduction asks for.

The resource implication can be separated from these analytic obligations. Both representations use width to resolve the same population predictor; response memory additionally uses order to resolve the learned history. The following conditional corollary makes the resulting saving precise while retaining the original dense Gaussian W_0 .

Corollary 3.5 (Conditional learned-state saving). *Fix the data, $L \geq 2$, d , a test probability measure Q and a common time interval I . Let f_n be the dense predictor and $\hat{f}_{n,q}$ the response-memory predictor coupled through the same initialization. Suppose there are constants $A, B > 0$ and thresholds n_0, q_0 , independent of n, q , such that for every $n \geq n_0$, $q \geq q_0$,*

$$\begin{aligned} \left(\mathbb{E} \sup_{t \in I} \|f_n(t) - f_\infty(t)\|_{L^2(Q)}^2 \right)^{1/2} &\leq A n^{-1/2}, \\ \left(\mathbb{E} \sup_{t \in I} \|\hat{f}_{n,q}(t) - f_n(t)\|_{L^2(Q)}^2 \right)^{1/2} &\leq B q^{-2}. \end{aligned} \tag{10}$$

Then population prediction error at most ε is certified by

$$n \asymp \varepsilon^{-2}, \quad q \asymp \varepsilon^{-1/2} \asymp n^{1/4}.$$

The least leading moving-state counts satisfying these error certificates are

$$M_{\text{dense}}(\varepsilon) \asymp (L-1)A^4\varepsilon^{-4}, \quad M_{\text{memory}}(\varepsilon) \asymp (L-1)mA^2\sqrt{B}\varepsilon^{-5/2}. \tag{11}$$

In particular,

$$\frac{M_{\text{memory}}(\varepsilon)}{M_{\text{dense}}(\varepsilon)} \asymp \frac{m\sqrt{B}}{A^2} \varepsilon^{3/2} \longrightarrow 0. \tag{12}$$

The same statement applies on $I = [0, \infty)$ if both estimates (10) hold there with finite constants and uniform thresholds.

Proof. The triangle inequality gives error at most $A/\sqrt{n} + B/q^2$. Allocate $4\varepsilon/5$ to width and $\varepsilon/5$ to history: take $n = \lceil 25A^2/(16\varepsilon^2) \rceil$ and $q = \lceil \sqrt{5B/\varepsilon} \rceil$, enlarged to their fixed thresholds. The moving state is $2(L-1)mnq + (d+1)n$, with an additional $O(q^2)$ for the joint clock. Its leading term is of order $\varepsilon^{-5/2}$. Dense storage instead takes $n = \lceil A^2/\varepsilon^2 \rceil$ and $(L-1)n^2 + (d+1)n$ moving entries. Optimizing the leading product nq gives exactly the stated 4:1 allocation; Appendix E.1 provides the calculation and constants. Dividing the counts proves (12). $\square$

Thus quadratic history accuracy saves a factor of order $\varepsilon^{-3/2}$ in evolving state relative to the dense certificate. The assumptions concern the same test-prediction metric and time interval; an assumed dense $O(\varepsilon^{-4})$ cost alone does not supply the second estimate in (10). These are costs certified by the stated bounds, not necessary costs for every dense implementation. The retained W_0 still occupies $\Theta(Ln^2)$ fixed entries and requires dense actions. The saving therefore concerns the learned dynamics. Appendix E.1 also gives the ε^{-3} cost under a q^{-1} estimate and the near- $\varepsilon^{-5/2}$ version when the quadratic rate has a subpolynomial correction. No all-time error hypothesis is inferred merely from small labels or from the finite-horizon theorems.

Appendix C records an earlier, weaker closure of the population flow itself, in which the learned middle layer is confined to a frozen span of initialization observables; it converges without a rate, and its limitations motivated the evolving memories of the main result.

4 Experiments

We integrate finite networks and compare response memory with dense gradient flow from the same initialization, using the learning-speed clock unless stated. We first examine the developing function at shared physical times, then compare fitted functions at each model’s first crossing of training MSE 0.001, and then ask whether reproducing that function requires the response-memory dynamics. Shared-time discrepancies address trajectory tracking, while individually fitted endpoints test whether the closure reproduces the function selected by dense training. The trajectory panel uses a fresh checkpoint replay; frozen-kernel controls are evaluated analytically, and the remaining experimental panels reuse saved predictions. No intermediate evolution is inferred from endpoints.

Learning in physical time. Figure 1 follows the paired-label circle task with two hidden tanh layers of width 2048. Dense training and orders $q = 1, 3, 7$ have predictions saved at the same 39 physical times between 0 and 80, with denser, approximately logarithmic spacing from 0.5 onward. Four snapshots show the function developing from the common initialization. Over all shared snapshots, on 2048 uniform circle queries against the finer dense replay, the RMS discrepancy stays below 0.071 for $q = 1$, 0.0059 for $q = 3$ and 0.00018 for $q = 7$; the full curves are in Figure 9 (Appendix H), together with the numerical sensitivity scale obtained by summing the dense and closure coarse/fine prediction changes, which the smallest discrepancies approach. The replay preserves the original initialization and uses float64 adaptive Heun integration at two tolerances for every model. These discrete observations illustrate tracking; they do not certify the error between observations.

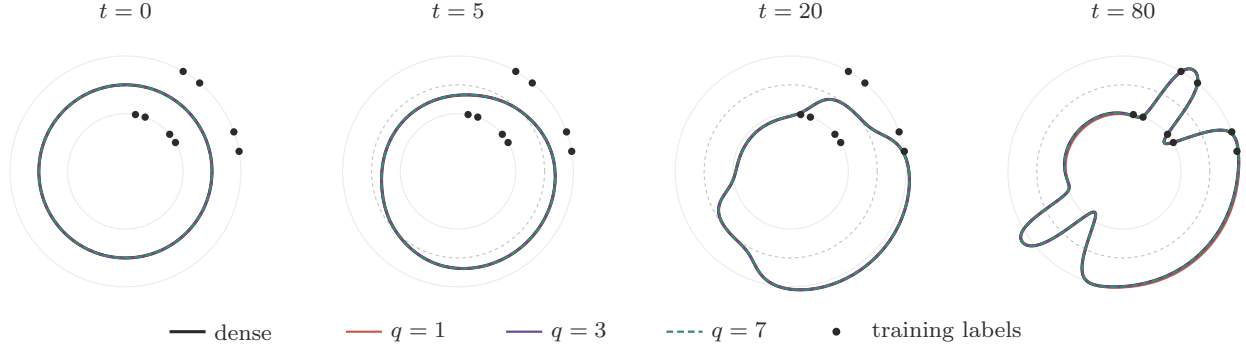


Figure 1: Learning at common physical times. Paired-label task, two hidden tanh layers, width 2048: saved functions of dense training and of response memory at orders $q = 1, 3, 7$ at the same physical times $t = 0, 5, 20, 80$. Radius is $3 + f(\theta)$, the dashed ring marks $f = 0$ and the light rings $f = \pm 1$; black dots are training labels. The RMS discrepancies at all 39 shared times are in Figure 9.

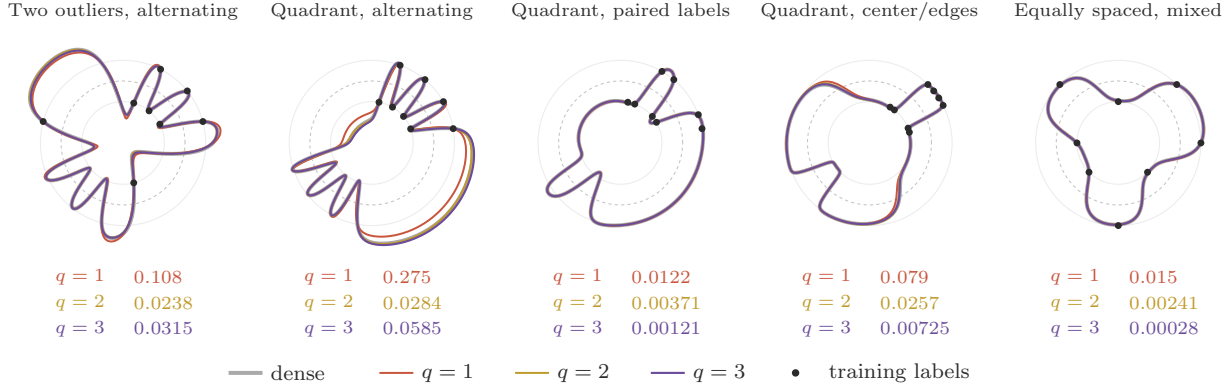


Figure 2: The fitted function across five tasks. Three hidden tanh layers, width 4096, and memory orders $q = 1, 2, 3$, at each model’s training-MSE 10^{-3} endpoint. Angle represents input direction; radius is $3 + f(\theta)$, the dashed ring marks $f = 0$ and the light rings $f = \pm 1$. The thick grey curve is dense training, coloured curves are response memory, and black dots are training labels. The numbers under each panel are RMS differences from dense on 8192 uniform circle points.

Fitted functions across tasks. With eight labelled points on a circle, many functions fit the labels, so agreement at training points says little. We compare the selected function on 8192 angles for five tasks with alternating labels, clustered points and large unlabelled arcs. Figure 2 displays the three-hidden-layer results; the two-hidden-layer gallery is in Appendix G. The same initialization is used within each task and architecture. A supplementary four-hidden-layer ReLU example on the sphere appears in Appendix F.

In two hidden layers (Figure 6, Table 2 in Appendix G), the fitted-function discrepancy falls by 1.5–3 orders of magnitude from $q = 1$ to $q = 7$ on every task. In three hidden layers, $q = 2$ and $q = 3$ have RMS differences below 0.06 on all five tasks, and below 0.01 on three. Order is not monotone at these small values: $q = 2$ beats $q = 3$ on both alternating tasks; the observed numerical sensitivity is two orders of magnitude below these differences, and no power law is fitted (Figure 10, Appendix H). The hidden features move substantially: the *absolute* activation RMS changes from initialization are 0.35–0.65, 0.38–0.74 and 0.46–0.68 in the three layers, across the 20 finest dense and closure runs. The moving state at $q = 3$ is about 3 MiB against 256 MiB for the

dense learned matrices; retained initialized matrices are additional.

Frozen dictionaries built at initialization are a different alternative: on the outlier task they reach RMS 0.55–1.38 with about 45 vectors, against 0.073 for 48 response moments. Together with the factor control that follows, this motivates both parts of the method: evolving response directions and an evolution law derived from the network’s own history.

Capturing the learned function requires the right dynamics. Figure 3 compares dense training with its full initialization-frozen empirical tangent kernel, directly trained factors, and response memory. All four fit the training labels to MSE approximately 0.001, yet select different functions between those labels. On the alternating-label quadrant task, the frozen kernel has whole-circle RMS discrepancy 171.03 from dense, the two factor seeds have discrepancies 0.6766 and 1.5413, and memory at $q = 3$ has discrepancy 0.04561. The kernel includes every parameter block with the canonical mobilities; its construction and all five task comparisons are given in Appendix I and figure 14.

Both response memory and directly trained factors retain the same $W_0^{(2)}$ and have evolving low-rank learned corrections. Their dynamics differ: ordinary Euclidean factor gradient flow does not induce canonical matrix gradient flow. Figure 12 summarizes the comparison across tasks, and Figure 13 shows the fitted functions on the paired-label task at correction rank bound 24. Memory has whole-circle RMS 0.00282 from dense, against 0.375 and 0.204 for the two factor seeds. Across all five tasks and three ranks, memory has smaller discrepancy in all 29 resolved fitted comparisons; one capped factor run is excluded. Twenty-eight wins exceed a factor of two. Together, the controls show that the reproduced predictor goes beyond the initial tangent model and this rank-matched factor flow. They support the response-memory evolution law, without asserting a universal comparison with kernels or low-rank algorithms.

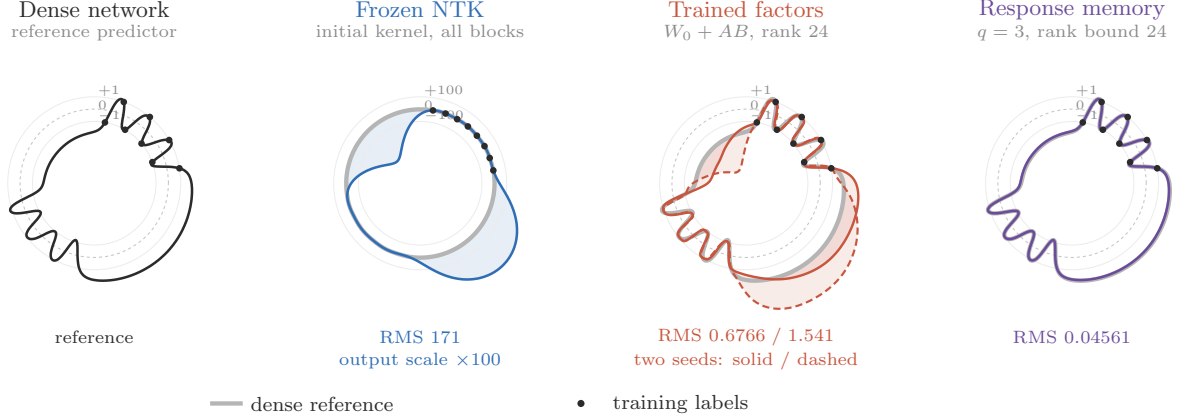
MNIST. With 100 training images and width 4096, held-out discrepancies are 0.00325, 0.00108 and 0.00120 for $q = 1, 2, 3$, retaining the small deterioration from $q = 2$ to $q = 3$; the predictions and their signed residuals are shown in Figure 11 (Appendix H). All models have identical test accuracy (94.7%). With 1000 training images and width 1024, the discrepancies are 0.019, 0.0019 and 0.0012, with 97.5% accuracy for dense, $q = 2$ and $q = 3$. Prediction fidelity and label accuracy are separate observables.

Two clocks. The joint clock’s better rate is asymptotic. At $q = 5$ it improved some cases (tanh alternating quadrant $0.0066 \rightarrow 0.0035$; GELU outliers $0.0020 \rightarrow 0.0010$) and was much worse on another (GELU alternating quadrant $0.013 \rightarrow 1.00$, reduced to 0.10 by halving the step). Over an 82-configuration sweep, on the 65 configurations where dense and all orders fit, the joint clock’s median discrepancies were 0.031, 0.013 and 0.0066 for $q = 1, 2, 3$.

5 Related work and comparison

We organize prior work by where it sits on the axis of the introduction: whether it targets the specified deep nonlinear feature-learning flow and, when it does, how its evolving state is represented. Table 1 places one representative work from each description on that axis: which of the three properties it keeps, what its order is, how its moving state scales (Section 2), and how its error decreases in the order. The entries are orders of magnitude, not a ranking: constants differ between rows and depend on the horizon. Section 3.6 states what the theorems of this paper give at fixed

Quadrant, alternating



Same initialized network; each model at training MSE $\simeq 0.001$, at its own fitting time.
RMS differences from dense use 8192 circle queries. Signed ticks give output values; any wider NTK scale is marked explicitly.

Figure 3: Fitting the labels does not determine the learned function. Alternating-label quadrant task, eight training inputs, two hidden tanh layers and width 2048. From left: dense training, the full frozen empirical NTK, directly trained factors at rank 24 (two seeds), and response memory at $q = 3$ with rank bound 24. All start from the same physical network and are evaluated at their own training-MSE 0.001 endpoints. Gray curves show the dense reference in each panel; black dots are training labels. Angle is input direction and radius is linear in output. The NTK panel uses a 100-times wider output scale, including its reference curve and labels; signed ticks show original output units. RMS values are unscaled discrepancies from dense over 8192 circle queries, not errors against unseen labels. The complete five-task comparison appears in Figure 14.

width, and Appendix E defines the error metric and the hypotheses under which the rows can be compared against the population flow.

Off the axis: descriptions that give up one property. Exact solutions of deep linear networks keep depth and feature learning but not nonlinearity (Saxe et al., 2014); their infinite-width limit is a gradient flow on ℓ^2 operators added to the frozen initialization (Chizat et al., 2024), the cleanest example of an operator description. Wasserstein and mean-field limits describe one-hidden-layer networks as the gradient flow of a measure over neurons (Chizat and Bach, 2018; Mei et al., 2018; Sirignano and Spiliopoulos, 2019); they keep nonlinear feature learning but not depth. The neural tangent kernel and lazy training keep depth and nonlinearity but freeze the features, reducing training to a fixed kernel (Jacot et al., 2018; Lee et al., 2019; Chizat et al., 2019). Each is simple because of what it sacrifices, and each is exact about a different object. These regime changes can leave a discrepancy that increasing width does not remove. This concerns the learning dynamics, not whether a shallow or kernel model can represent a particular fitted function.

On the axis, paying in history. Tensor Programs IV proves feature-learning limits at each fixed number of steps, with a recursion that includes variables indexed by every earlier step (Yang and Hu, 2021). Feature-learning DMFT evolves two-time activation, gradient and response kernels, with memory quadratic and time cubic in the number of time points (Bordelon and Pehlevan, 2022), and its finite-width extension adds perturbative fluctuations (Bordelon and Pehlevan, 2023). Direct two-time storage is smaller than dense $O(n^2)$ weight storage per layer when the number of samples times the number of time steps is well below the width. Runtime additionally depends on sampled

Table 1: Descriptions of training on a common state–accuracy axis, one representative work per row. Columns: nonlinearity effective in training, more than one hidden layer, feature learning at leading order; the order that buys accuracy; the moving state per hidden layer, with constant factors and fixed sources (the initialization W_0 , the NTK matrix, the top NTH tensor) omitted; and how the error decreases in that order, uniform on a fixed horizon. m samples, n width, d input dimension, q the order of each row’s own model. Tensor Programs are counted by the stored past responses and DMFT by its two-time kernels; the tangent hierarchy captures feature learning perturbatively, as an $n^{-1/2}$ correction in its own scaling, in which its rate is proved. The multilayer mean-field rate is conditional (Appendix E); the proved rate is n^{-c} with $c < \frac{1}{2}$. The three works above the line are exact about a different object and have no order that reaches ours.

	nonlin.	depth	features	order	moving state	error
Deep linear (Saxe et al., 2014)		✓	✓	—	d scalar modes	—
Two-layer mean field (Mei et al., 2018)	✓		✓	—	nd	—
Neural tangent kernel (Jacot et al., 2018)	✓	✓		—	m	$O(1)$ bias
Multilayer mean field (Nguyen and Pham, 2023)	✓	✓	✓	width n	n^2	$n^{-1/2}$
Tensor Programs (Yang and Hu, 2021)	✓	✓	✓	steps	$mn \times \text{steps}$	exact, no rate
DMFT (Bordelon and Pehlevan, 2022)	✓	✓	✓	steps	$m^2 \times \text{steps}^2$	step size
Tangent hierarchy (Huang and Yau, 2020)	✓	✓	(✓)	kernels q	$\sim m^{q-1}$	$n^{-q/2}$
Response memory, speed clock	✓	✓	✓	moments q	mnq	q^{-1} (Theorem 3.1)
Response memory, joint clock	✓	✓	✓	moments q	$mnq + q^2$	q^{-2} (Theorem 3.2)

single-site paths and self-consistency sweeps (Appendix E.3). Multilayer mean-field theory and Neural Feature Flow evolve fields on neuronal ensembles (Nguyen and Pham, 2023; Fang et al., 2020). These keep all three properties within their respective parameterizations. They compress width: DMFT, for instance, has kernel state independent of the number of neurons. Direct history solvers retain time-indexed state; field descriptions instead retain functions on population spaces. Their representation and approximation costs must be specified separately. Our result is complementary: it keeps the neurons and compresses time.

On the axis, paying in tuples of samples. The Neural Tangent Hierarchy (Huang and Yau, 2020) evolves the outputs together with the tangent kernels of order $2, \dots, q$, the j -th a function of j training points, and closes by freezing the top kernel. The hierarchy is an exact identity in any parametrization, and above order two it captures feature learning; the neural tangent kernel is its lowest truncation. What it pays is combinatorial in the samples: the moving state through order q is $\sum_{j=1}^{q-1} m^j \sim m^{q-1}$ scalars beneath a frozen tensor of m^q entries (a direct count rather than a statement of that work), against a state linear in m here. Its published guarantee uses width-suppressed higher kernels: for even q it gives an $n^{-q/2}$ factor multiplied by time-dependent terms, on a width-dependent horizon and under specified data and regularity assumptions. Those estimates cannot simply be transferred to the present scaling, where feature and kernel evolution can remain order one. The matched hierarchy still captures feature learning, but its truncation error must be controlled through higher-kernel growth (Appendix E.2). Response memory controls the full parameter trajectory at fixed width, with a state linear in q and an error decreasing polynomially in q .

Memory and low rank. HiPPO provides online Legendre projection of a given signal (Gu et al., 2020); we apply it to endogenous, coupled training histories and prove stability of the resulting feedback loop. Low-rank training methods constrain the learned increment to evolving factors (Zhao et al., 2023); response-memory factors are causal summaries of the actual forward and backward histories, and a directly trained factorization of matched rank performs far worse (Section 4).

Seen from far enough away, every description of training that keeps depth, nonlinearity and feature learning pays for the limit in some currency and buys accuracy with the matching knob (Table 1). The present contribution is that, at fixed width, the evolving learned state is linear in sample count and approximation order, with an error certified against the full trajectory (Section 3.6); whether the same holds against the population flow is exactly Conjecture 3.4. The exponents of Appendix E are the shape of that trade-off under stated hypotheses, not a verdict, and our constants have no proven growth law in the horizon or the width, so “constant in time” is a statement about the state, not about the constants.

6 Discussion: toward an idealized model of learning

We asked whether the idealized limit of deep feature learning can be described by a small state, in the way a large physical system is described by a few collective variables. Exact scalar closure encounters a concrete obstruction in the bounded-contraction class, while general exact population compression remains open. Approximately, a constructive answer is available. History-based limiting descriptions compress width; we keep the neurons and compress the history. Measured in a clock that follows learning, the memory of deep nonlinear feature learning is summarized by a few moments per neuron and sample, with an error that decreases polynomially in the number of moments and a state that does not grow as steps elapse within the chosen horizon. An illuminating analogue is the moment closure of kinetic theory: a finite collection of moments represents a much larger dynamical object. Here the history moments evolve autonomously, and the feedback of their truncation error is controlled along the full parameter trajectory.

The variables that survive are *neuron-level memory variables*: for each neuron and sample, compressed summaries of how it has responded forward and backward, coupled through a fixed random operator. They become collective only at the population level, and Conjecture 3.4 states what that would require. Other gaps remain. The initialized random operator W_0 is kept exactly; it plays the role of a fixed random environment, and compressing it while preserving its forward and adjoint reuse is the main open problem. The constants depend on width and horizon, and the small orders that work in practice are not explained by the asymptotic rates. We have not ablated the clock against physical time. Finally, the theory covers smooth activations, and the memory grows with the number of training samples, so learning over a continuum of inputs needs a further representation of the input dependence.

References

- Blake Bordelon and Cengiz Pehlevan. Self-consistent dynamical field theory of kernel evolution in wide neural networks, 2022. URL <https://arxiv.org/abs/2205.09653>.
- Blake Bordelon and Cengiz Pehlevan. Dynamics of finite width kernel and prediction fluctuations in mean field neural networks, 2023. URL <https://arxiv.org/abs/2304.03408>.
- Lénaïc Chizat and Francis Bach. On the global convergence of gradient descent for over-parameterized models using optimal transport. In *Advances in Neural Information Processing Systems*, volume 31, 2018.
- Lénaïc Chizat, Maria Colombo, Xavier Fernández-Real, and Alessio Figalli. Infinite-width limit of deep linear neural networks. *Communications on Pure and Applied Mathematics*, 2024. doi: 10.1002/cpa.22200.

- Lénaïc Chizat, Edouard Oyallon, and Francis Bach. On lazy training in differentiable programming. In *Advances in Neural Information Processing Systems*, volume 32, 2019.
- Cong Fang, Jason D. Lee, Pengkun Yang, and Tong Zhang. Modeling from features: A mean-field framework for over-parameterized deep neural networks, 2020. URL <https://arxiv.org/abs/2007.01452>.
- Albert Gu, Tri Dao, Stefano Ermon, Atri Rudra, and Christopher Ré. HiPPO: Recurrent memory with optimal polynomial projections, 2020. URL <https://arxiv.org/abs/2008.07669>.
- Jiaoyang Huang and Horng-Tzer Yau. Dynamics of deep neural networks and neural tangent hierarchy, 2020. URL <https://arxiv.org/abs/1909.08156>.
- Arthur Jacot, Franck Gabriel, and Clément Hongler. Neural tangent kernel: Convergence and generalization in neural networks, 2018. URL <https://arxiv.org/abs/1806.07572>.
- Jaehoon Lee, Lechao Xiao, Samuel S. Schoenholz, Yasaman Bahri, Roman Novak, Jascha Sohl-Dickstein, and Jeffrey Pennington. Wide neural networks of any depth evolve as linear models under gradient descent, 2019. URL <https://arxiv.org/abs/1902.06720>.
- Song Mei, Andrea Montanari, and Phan-Minh Nguyen. A mean field view of the landscape of two-layer neural networks, 2018. URL <https://arxiv.org/abs/1804.06561>.
- Phan-Minh Nguyen and Huy Tuan Pham. A rigorous framework for the mean field limit of multilayer neural networks. *Mathematical Statistics and Learning*, 6(3/4):201–357, 2023. doi: 10.4171/MSL/42.
- Andrew M. Saxe, James L. McClelland, and Surya Ganguli. Exact solutions to the nonlinear dynamics of learning in deep linear neural networks. In *International Conference on Learning Representations*, 2014.
- Justin Sirignano and Konstantinos Spiliopoulos. Mean field analysis of neural networks: A law of large numbers, 2019. URL <https://arxiv.org/abs/1805.01053>.
- Greg Yang and Edward J. Hu. Tensor programs IV: Feature learning in infinite-width neural networks. In *Proceedings of the 38th International Conference on Machine Learning*, volume 139 of *Proceedings of Machine Learning Research*, pages 11727–11737. PMLR, 2021. URL <https://proceedings.mlr.press/v139/yang21c.html>.
- Jiawei Zhao, Yifei Zhang, Beidi Chen, Florian Schäfer, and Anima Anandkumar. InRank: Incremental low-rank learning, 2023.

A Proof of Theorem 3.1

Throughout the appendices we write $b_a^{(\ell)} = r_a \delta_a^{(\ell)} / \rho$ for the residual-weighted backward response, $h_a^{(\ell-1)}(\xi)$ and $b_a^{(\ell)}(\xi)$ for the forward and backward histories as functions of the clock coordinate ξ (extended over the unit prefix as in the main text), $\bar{h}_a^{(\ell-1)}$ and $\bar{\delta}_a^{(\ell)}$ for their moment matrices as in the main text, and $h_a^{(\ell-1)*}$, $b_a^{(\ell)*}$ for the values at the current endpoint of the degree- $(q-1)$ Legendre projections of the histories. The only approximation-theoretic input is the following Legendre tail bound.

Lemma A.1 (Legendre tail). *For $u \in H^1(0, \tau; \mathbb{R}^N)$,*

$$\|u - \Pi_q u\|_{L^2(0, \tau)}^2 \leq \frac{1}{q(q+1)} \int_0^\tau \xi(\tau - \xi) \|u'(\xi)\|^2 d\xi \leq \frac{\tau^2}{4q(q+1)} \|u'\|_{L^2(0, \tau)}^2. \quad (13)$$

Proof. On $[0, 1]$, the shifted Legendre equation is

$$-\frac{d}{dx}\{x(1-x)p'_k(x)\} = k(k+1)p_k(x).$$

Integration by parts shows that the derivatives are orthogonal in the weight $x(1-x)$ and have squared norm $k(k+1)/(2k+1)$. Applying Bessel's inequality to u' in this weighted space gives

$$\sum_{k \geq 1} k(k+1) \frac{\|c_k\|^2}{2k+1} \leq \int_0^1 x(1-x) \|u'(x)\|^2 dx,$$

where $c_k = (2k+1) \int_0^1 u(x)p_k(x) dx$. Parseval, polynomial density and $k(k+1) \geq q(q+1)$ for $k \geq q$ give the first bound on $[0, 1]$. Rescaling and $\xi(\tau - \xi) \leq \tau^2/4$ give (13); summing componentwise proves the vector statement. $\square$

Step 1: dense existence and a common compact region. The mobility matrix $\mathcal{D}$ has largest eigenvalue n , and $F = -\mathcal{D}\nabla\mathcal{L}$. Hence

$$\dot{\mathcal{L}} = -\left\|\mathcal{D}^{-1/2}\dot{\theta}\right\|^2, \quad \int_0^t \left\|\dot{\theta}(s)\right\|^2 ds \leq n\rho(0)^2. \quad (14)$$

Thus $\|\theta(t) - \theta(0)\| \leq \sqrt{nt}\rho(0)$. A finite maximal endpoint would have a finite limiting state, from which local existence would extend the solution, so the dense flow is global.

Now fix T and let

$$R = \|\theta(0)\| + \sqrt{nT}\rho(0) + 1, \quad \mathcal{B} = \{\vartheta : \|\vartheta\| \leq R\}.$$

The dense path stays at distance at least one from $\partial\mathcal{B}$. We stop the closure at its first exit from $\mathcal{B}$; the rest of the proof uses constants on this explicit ball and eventually rules out that exit.

To make the constants concrete, set $X = \max_a \|x_a\|/\sqrt{d}$ and $\alpha_0 = X$, and choose recursively

$$\begin{aligned} s_\ell &= \sup_{|z| \leq R\alpha_{\ell-1}} |\phi'(z)|, & \alpha_\ell &= \sqrt{n} \sup_{|z| \leq R\alpha_{\ell-1}} |\phi(z)|, \\ \beta_L &= Rs_L, & \beta_\ell &= s_\ell R\beta_{\ell+1} \quad (\ell < L). \end{aligned} \quad (15)$$

Then $\|h_a^{(\ell)}\| \leq \alpha_\ell$ and $\|\delta_a^{(\ell)}\| \leq \beta_\ell$ on $\mathcal{B}$. Put

$$\begin{aligned} Y &= \left(m^{-1} \sum_a y_a^2\right)^{1/2}, & \varrho &= R\alpha_L/n + Y, & S &= T\varrho, & A &= 1 + S, \\ v_1 &= 2\beta_1 X, & v_\ell &= 2\beta_\ell \alpha_{\ell-1}/n, & v_w &= 2\alpha_L, \\ V &= \left(v_1^2 + \sum_{\ell=2}^L v_\ell^2 + v_w^2\right)^{1/2}. \end{aligned} \quad (16)$$

On the stopped interval, $\rho \leq \varrho$, $\tau \leq A$ and $\|F\| \leq V\rho$. Since F is locally Lipschitz, it has some finite Lipschitz constant Λ_F on $\mathcal{B}$.

Step 2: an exact projection-error energy. For any single forward or backward history u , let $D_u(t) = \|u - \Pi_q u\|_{L^2(0, \tau(t))}^2$, and let u^* be the polynomial projection evaluated at the current endpoint. Differentiating the raw energy and the projected energy with (6), every basis-dilation term cancels, leaving

$$\dot{D}_u = \rho \|u - u^*\|^2, \quad D_u(t) = \int_0^t \rho(s) \|u(s) - u^*(s)\|^2 ds. \quad (17)$$

The jump at the end of the backward prefix is harmless, since only square integrability and growing integrals are used.

Next differentiate (7). The physical closure equation is exactly

$$\dot{\hat{\theta}}_q = F(\hat{\theta}_q) + E, \quad E_1 = E_w = 0, \quad (18)$$

with internal defects

$$E_\ell = \frac{2\rho}{nm} \sum_a (b_a^{(\ell)} - b_a^{(\ell)*})(h_a^{(\ell-1)} - h_a^{(\ell-1)*})^\top. \quad (19)$$

The two mixed projection terms vanish by orthogonality, so the defect is a pure product of errors. Consequently

$$\int_0^t \|E_\ell(s)\|_F ds \leq \frac{2}{nm} \sum_a \sqrt{D_{b,\ell,a}(t) D_{h,\ell,a}(t)}. \quad (20)$$

This inequality is the bridge from a same-history approximation bound to a trajectory theorem. Define also the proof-only accumulator

$$W_{\text{acc}}^{(\ell)}(t) = W_0^{(\ell)} - \frac{2}{nm} \sum_a \int_0^t \hat{r}_a \hat{\delta}_a^{(\ell)} (\hat{h}_a^{(\ell-1)})^\top ds, \quad R_\ell = \widehat{W}^{(\ell)} - W_{\text{acc}}^{(\ell)}.$$

Then $R_\ell(0) = 0$ and $\dot{R}_\ell = E_\ell$, so (20) controls the accumulated absolute velocity error, not only a signed endpoint integral.

Step 3: forward histories have a q -independent derivative bound. Before changing variables, note that a nonstationary learning-speed solution cannot first reach $\rho = 0$ at a finite regular state. Every raw velocity vanishes there, and local uniqueness applied backward from that equilibrium would force the preceding path to be stationary. Hence $\xi = \tau(t)$ is a valid coordinate on every nonstationary compact segment; the raw algorithm itself remains defined at $\rho = 0$ without division.

Write primes for derivatives in $\xi = \tau(t)$ and define

$$Z_\ell(t) = \frac{1}{m} \sum_a \int_0^{\tau(t)} \left\| (h_a^{(\ell)})'(\xi) \right\|^2 d\xi.$$

The first layer satisfies $\left\| (h_a^{(1)})' \right\| \leq s_1 v_1 X$. For later layers, the chain rule gives

$$(h_a^{(\ell)})' = \phi'(z_a^{(\ell)}) \odot \left[(F_\ell/\rho + E_\ell/\rho) h_a^{(\ell-1)} + \widehat{W}^{(\ell)} (h_a^{(\ell-1)})' \right]. \quad (21)$$

The delicate term is the defect, because endpoint evaluation of a polynomial projection can amplify errors by a factor that grows with q . Endpoint evaluation of a degree- $(q-1)$ polynomial has

operator norm $q/\sqrt{\tau}$ from $L^2(0, \tau)$, because $\sum_{k < q} (2k + 1) = q^2$. Projection contraction and $m^{-1} \sum_a \left\| b_a^{(\ell)} \right\|^2 \leq \beta_\ell^2$ then imply

$$\left(m^{-1} \sum_a \left\| b_a^{(\ell)*} \right\|^2 \right)^{1/2} \leq q\beta_\ell, \quad \left(m^{-1} \sum_a \left\| b_a^{(\ell)} - b_a^{(\ell)*} \right\|^2 \right)^{1/2} \leq (q+1)\beta_\ell.$$

Combining this with Lemma A.1, (17) and (19) yields

$$\int_0^t \rho \|E_\ell / \rho\|_F^2 ds \leq \frac{2\beta_\ell^2 A^2}{n^2} Z_{\ell-1}(t). \quad (22)$$

Here the endpoint factor $q+1$ cancels the q^{-1} projection factor. This cancellation is the step that prevents a loss of convergence order with depth.

Define finite, q -independent bounds recursively by

$$\begin{aligned} \bar{Z}_1 &= S(s_1 v_1 X)^2, \\ \bar{Z}_\ell &= 3s_\ell^2 \left[S v_\ell^2 \alpha_{\ell-1}^2 + \left(R^2 + \frac{2\beta_\ell^2 A^2 \alpha_{\ell-1}^2}{n^2} \right) \bar{Z}_{\ell-1} \right]. \end{aligned} \quad (23)$$

Integrating (21), and using $\|x + y + z\|^2 \leq 3(\|x\|^2 + \|y\|^2 + \|z\|^2)$ together with (22), proves inductively that $Z_\ell \leq \bar{Z}_\ell$.

Step 4: accumulated defect and feedback. The zero backward prefix gives $m^{-1} \sum_a D_{b,\ell,a} \leq S\beta_\ell^2$, and Lemma A.1 gives

$$\frac{1}{m} \sum_a D_{h,\ell,a} \leq \frac{A^2 \bar{Z}_{\ell-1}}{4q(q+1)}.$$

Summing (20) over links, we obtain

$$\int_0^t \|E(s)\| ds \leq \frac{B_{\text{LS}}}{\sqrt{q(q+1)}}, \quad B_{\text{LS}} = \frac{A}{n} \sum_{\ell=2}^L \beta_\ell \sqrt{S \bar{Z}_{\ell-1}}. \quad (24)$$

It remains to account for feedback. Subtracting the dense flow from the perturbed flow and applying the Lipschitz bound on F gives

$$e(t) \leq \frac{B_{\text{LS}}}{\sqrt{q(q+1)}} + \Lambda_F \int_0^t e(s) ds,$$

and Gronwall's inequality yields

$$\sup_{t \leq T} e(t) \leq \frac{B_{\text{LS}} e^{\Lambda_F T}}{\sqrt{q(q+1)}}. \quad (25)$$

Choose q_0 so that the right side is less than $1/2$. The closure then cannot reach $\partial \mathcal{B}$. Its moment representations, $1 \leq \tau \leq A$ and the response bounds place its whole finite state in a compact subset of the locally Lipschitz domain. A finite maximal endpoint would therefore extend, so the closure exists through T . Lipschitz dependence of the network responses on θ gives the final prediction statement.

Globally bounded activations. Suppose now that every activation and its first derivative are globally bounded. Set

$$\alpha_\ell = \sqrt{n} \|\phi\|_\infty, \quad s_\ell = \|\phi'\|_\infty, \quad B_w = \sqrt{\|w(0)\|^2 + nY^2T}.$$

For both the dense and the closure flows, the unchanged readout equation obeys

$$\frac{d}{dt} \|w\|^2 = nY^2 - 4n \frac{1}{m} \sum_a (f_a - y_a/2)^2 \leq nY^2.$$

Therefore $\|w\| \leq B_w$, $\rho \leq \varrho := B_w \alpha_L / n + Y$, $S = T\varrho$ and $\tau \leq A := 1 + S$. Start with $\beta_L = B_w s_L$ and proceed downward: for $\ell = L, \dots, 2$, define

$$D_\ell = \left\| W_0^{(\ell)} \right\|_F + \frac{2A\beta_\ell \alpha_{\ell-1}}{n}, \quad \beta_{\ell-1} = s_{\ell-1} D_\ell \beta_\ell.$$

Projection contraction in (8) gives $\left\| \widehat{W}^{(\ell)} \right\|_F \leq D_\ell$, and the backward recurrence then gives $\|\delta^{(\ell-1)}\| \leq \beta_{\ell-1}$. There is no circularity, because the descent starts from the bounded readout. Finally,

$$\left\| W^{(1)}(t) \right\|_F \leq \left\| W^{(1)}(0) \right\|_F + 2SX\beta_1.$$

These q -independent physical bounds, the moment integral representations and $1 \leq \tau \leq A$ place the order- q state in a compact set for every fixed q , so the closure continues through T for every $q \geq 1$. Repeating Steps 2–4 with these global constants gives (25) for every q . This completes the proof. $\square$

B The joint feature-learning clock: construction and proof of Theorem 3.2

Construction. Let Ψ be the concatenation over $\ell \geq 2$ and a of the forward and residual-weighted backward responses $(h_a^{(\ell-1)}, b_a^{(\ell)})$, so that the clock (9) reads $\dot{\tau} = g = \rho + \left\| \dot{\Psi} \right\|_2$ and $\|\mathrm{d}\Psi/\mathrm{d}\tau\| \leq 1$. History is placed at $\xi = \tau(s)$ with mass $\rho(s) \mathrm{d}s$, while the prefix $[0, 1]$ carries Lebesgue mass and constant initial responses; the resulting measure μ_t satisfies $\mathrm{d}\mu_t \leq \mathrm{d}\xi$. With $p = (p_0, \dots, p_{q-1})^\top$ evaluated at ξ/τ , $e = (1, \dots, 1)^\top$, and the triangular matrix $\mathsf{T}_{kk} = k$, $\mathsf{T}_{kj} = 2j + 1$ for $j < k$ (zero above the diagonal), the state consists of the moment matrices $\bar{h}_a^{(\ell-1)} = \int h_a^{(\ell-1)}(\xi) p^\top \mathrm{d}\mu_t$, $\bar{\delta}_a^{(\ell)} = \int b_a^{(\ell)}(\xi) p^\top \mathrm{d}\mu_t$ and one shared Gram matrix $G = \int pp^\top \mathrm{d}\mu_t$, which evolve by

$$\begin{aligned} \widehat{W}^{(\ell)} &= W_0^{(\ell)} - \frac{2}{nm} \sum_a (\bar{\delta}_a^{(\ell)} G^{-1} (\bar{h}_a^{(\ell-1)})^\top - b_a^{(\ell)}(0) h_a^{(\ell-1)}(0)^\top), \\ \dot{\bar{h}}_a^{(\ell-1)} &= \rho h_a^{(\ell-1)} e^\top - (g/\tau) \bar{h}_a^{(\ell-1)} \mathsf{T}^\top, \quad \dot{\bar{\delta}}_a^{(\ell)} = r_a \delta_a^{(\ell)} e^\top - (g/\tau) \bar{\delta}_a^{(\ell)} \mathsf{T}^\top, \\ \dot{G} &= \rho ee^\top - (g/\tau)(\mathsf{T}G + G\mathsf{T}^\top), \end{aligned} \tag{26}$$

with $G(0) = \text{diag}(1, \frac{1}{3}, \dots, \frac{1}{2q-1})$, only the zeroth moment columns nonzero initially, and the prefix product subtracted so that $\widehat{W}^{(\ell)}(0) = W_0^{(\ell)}$. With $h_a^{(\ell-1)*} = \bar{h}_a^{(\ell-1)} G^{-1} e$ and $b_a^{(\ell)*} = \bar{\delta}_a^{(\ell)} G^{-1} e$, differentiating the reconstruction cancels every g/τ term and gives the defect $E_\ell = \frac{2\rho}{nm} \sum_a (b_a^{(\ell)} - b_a^{(\ell)*})(h_a^{(\ell-1)} - h_a^{(\ell-1)*})^\top$, of the same product form as (19).

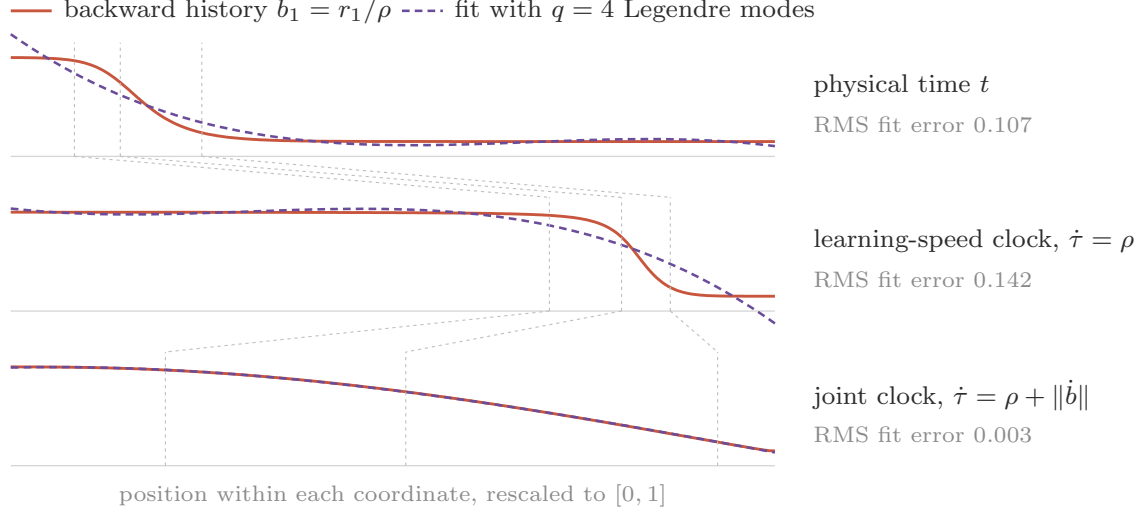


Figure 4: Why the clock matters, on a toy signal. Two residuals decay at different rates, $r_1 = e^{-4t}$ and $r_2 = 0.05 e^{-t/2}$, so the normalized backward signal $b = r/\rho$ switches direction late, when the loss is already small. Its first component is drawn in physical time, in the learning-speed clock and in the joint clock, each rescaled to $[0, 1]$, with a least-squares fit by four Legendre modes; dotted lines join the same physical instants. The learning-speed clock compresses the late phase and sharpens the switch; the joint clock spreads it out. The three fit errors are measured in three different coordinates, so they illustrate how smooth the history is in each, not a common-metric comparison of closure errors; the joint closure itself uses the residual-weighted projection with a Gram matrix.

Step 1: a lower bound on the dense residual. Assume $\rho_0 > 0$ as stated, and use the compact physical ball $\mathcal{B}$ from Appendix A. Retain the constants $\alpha_\ell, \beta_\ell, \varrho, S, \mathbf{A}, V, \Lambda_F$ from the proof of Theorem 3.1. On this ball the output map has a finite Lipschitz constant c_f : prediction and residual RMS change by at most $c_f \|\theta - \theta'\|$. Along dense flow,

$$|\dot{\rho}_{\text{dense}}| \leq c_f \|F(\theta)\| \leq c_f V \rho_{\text{dense}}.$$

Consequently

$$\rho_{\text{dense}}(t) \geq \mu := \rho_0 e^{-c_f V T} > 0, \quad 0 \leq t \leq T. \quad (27)$$

Step 2: an exact weighted projection energy. Consider the maximal regular closure interval stopped before either its physical state leaves $\mathcal{B}$ or its residual reaches $\mu/2$. We first derive a clock bound on this stopped interval without assuming small projection error. For any inserted vector history f , let

$$D_f = \|f - \Pi_t f\|_{L^2(\mu_t)}^2 = \int \|f\|^2 d\mu_t - \text{tr}(B_f G^{-1} B_f^\top).$$

Differentiating $G^{-1}G = I$ gives

$$\frac{d}{dt} G^{-1} = -\rho G^{-1} e e^\top G^{-1} + (g/\tau)(G^{-1} \mathbb{T} + \mathbb{T}^\top G^{-1}).$$

Together with (26), this cancels every dilation term and yields

$$\dot{D}_f = \rho \|f - f^*\|^2, \quad D_f(t) = \int_0^t \rho \|f - f^*\|^2 ds. \quad (28)$$

The initial error is zero because every prefix is constant. Thus (20) holds with the weighted errors in (28).

Step 3: order-independent coarse bounds. Put

$$Q = m \sum_{\ell=2}^L (\alpha_{\ell-1}^2 + \beta_{\ell}^2), \quad C_0 = \frac{AQ}{nm}.$$

The total mass of μ_t is at most $A = 1 + S$. Projection error is no larger than raw history energy, including the matching prefixes. Applying $2\sqrt{uv} \leq u + v$ in the defect estimate gives the order-independent bounds

$$\int_0^t \|E\| \, ds \leq C_0, \quad \int_0^t \|\dot{\hat{\theta}}\| \, ds \leq VS + C_0. \quad (29)$$

The second inequality uses $\dot{\hat{\theta}} = F(\hat{\theta}) + E$ and $\int_0^t \|F\| \, ds \leq V \int_0^t \rho \, ds \leq VS$. This total-variation estimate is stronger than bounded physical position, and it is obtained before any clock bound.

Step 4: a clock bound without a cap. It remains to control the derivative of the normalized response map. Let $\tilde{r} = r/\rho \in \mathbb{R}^m$. Then

$$\|\tilde{r}\|^2 = m, \quad D\tilde{r}[v] = \left(I - \frac{\tilde{r}\tilde{r}^\top}{m} \right) \frac{Dr[v]}{\rho}. \quad (30)$$

The matrix in parentheses is an orthogonal projector, and $\rho \geq \mu/2$ on the stopped interval. Differentiating the forward and backward recurrences uses only bounded physical weights and bounded first two activation derivatives on compact preactivation intervals; the first layer uses only ϕ' . Hence there is a finite, q -independent constant J such that

$$\|D\Psi(\vartheta)\|_{\text{op}} \leq J$$

throughout the stopped tube. The stated local smoothness also makes this derivative locally Lipschitz. Since $\dot{\Psi} = D\Psi(\hat{\theta})\dot{\hat{\theta}}$, (29) now gives

$$\tau_q(t) = 1 + \int_0^t (\rho + \|\dot{\Psi}\|) \, ds \leq A + J(VS + C_0) =: \Lambda_T. \quad (31)$$

No clock cap has been assumed or inserted.

Step 5: the sharp joint approximation estimate. We can now use the sharper approximation estimate. The entire concatenated history $\overline{\Psi}(\xi)$ is constant on the prefix, continuous at its join and, on the physical history,

$$\|\partial_\xi \overline{\Psi}\| = \frac{\|\dot{\Psi}\|}{\rho + \|\dot{\Psi}\|} \leq 1.$$

Therefore $\int_0^{\tau_q} \|\overline{\Psi}'\|^2 \, d\xi \leq \tau_q - 1$. The weighted projection is a best approximation in $L^2(\mu_t)$, and $d\mu_t \leq d\xi$, so comparison with the ordinary Legendre projection and Lemma A.1 give the joint bound

$$\sum_{\ell,a} (D_{h,\ell,a} + D_{b,\ell,a}) \leq \frac{\tau_q^2(\tau_q - 1)}{4q(q+1)} \leq \frac{\Lambda_T^2(\Lambda_T - 1)}{4q(q+1)}. \quad (32)$$

This is one estimate for the concatenated path; it does not introduce a separate dimension factor per unit or layer.

Step 6: feedback and continuation. Combining (20), (31), (32) and $2\sqrt{uv} \leq u + v$ yields

$$\int_0^t \|E\| \, ds \leq \frac{\Gamma_{\text{JF}}}{q(q+1)}, \quad \Gamma_{\text{JF}} = \frac{\Lambda_T^2(\Lambda_T - 1)}{4nm}. \quad (33)$$

Subtracting the dense flow and applying the Lipschitz constant Λ_F and Gronwall's inequality gives

$$\left\| \hat{\theta}_q(t) - \theta(t) \right\| \leq \frac{\Gamma_{\text{JF}} e^{\Lambda_F T}}{q(q+1)}. \quad (34)$$

Choose $q_0(T)$ so that, for every $q \geq q_0(T)$,

$$\frac{\Gamma_{\text{JF}} e^{\Lambda_F T}}{q(q+1)} \leq \frac{1}{2}, \quad c_f \frac{\Gamma_{\text{JF}} e^{\Lambda_F T}}{q(q+1)} \leq \frac{\mu}{4}. \quad (35)$$

The first inequality rules out physical exit, because the dense path stays at distance one from $\partial\mathcal{B}$. The second, together with (27), gives $\hat{\rho} \geq 3\mu/4 > \mu/2$, ruling out residual exit.

Finally, the matching prefix supplies, at every fixed q ,

$$G(t) \succeq \int_0^1 p(\xi/\tau_q(t)) p(\xi/\tau_q(t))^\top \, d\xi. \quad (36)$$

The right side is positive definite, since a zero quadratic form would make a nonzero polynomial vanish on an interval. Its entries vary continuously with τ_q , so on $1 \leq \tau_q \leq \Lambda_T$ its least eigenvalue has a positive minimum $\gamma_{q,T}$; this minimum need not be uniform in q . The moment integral representations, finite history mass, bounded responses and bounded fixed-order polynomials then bound every raw coordinate. Together with the physical, residual, clock and Gram bounds, the full state lies in a compact subset of its locally Lipschitz domain. A finite maximal endpoint therefore extends, which proves continuation through T and completes the theorem, with implied constant $\Gamma_{\text{JF}} e^{\Lambda_F T}$. $\square$

Implementation of the joint clock

Equation (9) appears implicit only because $\dot{\Psi}$ is written compactly; no nonlinear solve for g is needed. The internal velocities do not depend on g , so they can be computed first. Begin with $G^{-1}e$, the endpoint fits h^*, b^* , and the internal velocities

$$V_\ell = \dot{\widehat{W}}^{(\ell)} = -\frac{2\rho}{nm} \sum_a \left[b_a^{(\ell)} (h_a^{(\ell-1)*})^\top + b_a^{(\ell)*} (h_a^{(\ell-1)} - h_a^{(\ell-1)*})^\top \right]. \quad (37)$$

These expressions contain no g . Next compute $V_1 = F_1$ and $V_w = F_w$, and run a directional forward pass,

$$\dot{z}_a^{(1)} = V_1 x_a / \sqrt{d}, \quad \dot{z}_a^{(\ell)} = V_\ell h_a^{(\ell-1)} + \widehat{W}^{(\ell)} \dot{h}_a^{(\ell-1)}, \quad \dot{h}_a^{(\ell)} = \phi'(z_a^{(\ell)}) \odot \dot{z}_a^{(\ell)},$$

which gives $\dot{r}$ and $\dot{\rho}$. A reverse directional pass then differentiates the backward recurrence, using ϕ'' for $\ell \geq 2$, and yields

$$\dot{b}_a^{(\ell)} = \frac{\dot{r}_a \delta_a^{(\ell)} + r_a \dot{\delta}_a^{(\ell)}}{\rho} - b_a^{(\ell)} \frac{\dot{\rho}}{\rho}.$$

Only then do we evaluate $g = \rho + \left\| \dot{\Psi} \right\|$ and, finally, (26). Matrix and transpose actions use the same reconstructed operator. In practice one factors G and solves linear systems; an explicit inverse is unnecessary.

The joint clock is more regular in approximation coordinates, but it is not automatically better at low order. At $q = 1$ the polynomial space consists of constants, so changing the coordinate alone leaves the physical constant-mode closure unchanged. Higher orders pay for Gram solves and derivative passes, and the theoretical constants can be large.

C A frozen-span closure of the population flow

This appendix records a closure of the population flow itself, for the smallest case allowed by the contract: two hidden layers, $\phi = \tanh$, inputs on a circle ($d = 2$, $x/\sqrt{d} \in S^1$) with labels $|y| \leq Y$. The population state is the first-layer weight row $W^{(1)} \in L^2(\Omega_1; \mathbb{R}^2)$, the readout $w \in L^2(\Omega_2)$ and the middle operator $W^{(2)} = W_0^{(2)} + (W^{(2)} - W_0^{(2)})$, where $W_0^{(2)}$ is the initialized Gaussian operator and the learned part is Hilbert–Schmidt. The flow is (2) with $\frac{1}{m} \sum_a$ replaced by integration against a training law μ on inputs and labels, and the reverse action is always the true adjoint $(W^{(2)})^*$. We work on $0 \leq t \leq T = 1/200$ and on a Wasserstein neighborhood of the two-point task $\nu_* = \frac{1}{2}\delta_{(\sqrt{2}e_1, +1)} + \frac{1}{2}\delta_{(\sqrt{2}e_2, -1)}$; the neighborhood contains perturbed, nonorthogonal, nonatomic and noisy input laws, and neither its radius nor the interval shrinks with the closure order.

The compressed state. Enumerate the finite expressions built from constants, the Gaussian initial weights, rational affine combinations, $\sin$, $\cos$, ϕ , bounded products, and the initialized actions $W_0^{(2)}$ and $(W_0^{(2)})^*$. At order N this gives, on each population, a finite vector of *observables of the initialization*, which we normalize by a ridge-regularized Gram matrix so that they have unit scale. The saved state is then

- the joint law, on population 1, of the current first-layer weights $W^{(1)}$, their initial values, and the N observables;
- the joint law, on population 2, of the current readout w and the N observables;
- one finite matrix M representing the middle-layer action $W^{(2)}$ between the two observable vectors, initialized from $W_0^{(2)}$.

The closure evolves this state by the population flow itself, with $W^{(2)}$ replaced by the finite operator “project onto population-1 observables, apply M , expand in population-2 observables” and $(W^{(2)})^*$ by its transpose (details below). It is autonomous, uses only the current state and the training law, and is restartable from its own saved state. Initial and current values are stored jointly, so same-neuron quantities such as feature displacement are genuine observables of the state.

Theorem C.1 (Autonomous population closure). *For every separately fixed training law in the stated neighborhood of ν_* and every order N , the closure is uniquely well posed on $[0, T]$ and restartable from its saved state, and as $N \rightarrow \infty$ its prediction f_N converges to the population prediction f uniformly on $[0, T]$ and over the whole input circle. Every separately fixed same-population joint observation—initial and current hidden pairs, feature displacements, forward and adjoint actions, bounded nonlinear pushforwards and quadratic contractions—converges uniformly on $[0, T]$ in Wasserstein distance W_2 .*

The proof (below) controls the distance between closure and population flow through the error of the finite observable basis; unbounded backward fields force a truncation that turns Gronwall’s inequality into an Osgood inequality, whose solution still tends to zero with the basis error.

Proposition C.2 (Both layers learn, nonlinearly). *On a Wasserstein neighborhood of ν_* there is a common time at which both hidden populations have strictly positive mean-square displacement of their activations from initialization, and strictly positive nonaffinity $\inf_{\alpha, \beta} \mathbb{E}|\phi(z) - \alpha z - \beta|^2$ of their preactivations.*

The small-time expansion of the flow at ν_* (below) shows that both hidden displacements are nonzero at order t^2 , a Gaussian preactivation has positive nonaffinity, and continuity extends both properties to a neighborhood. The closure therefore converges on a family where all three properties of the introduction hold.

What this compresses, and why it is not the main result. The learned operator $W^{(2)} - W_0^{(2)}$ of the population flow is replaced by a finite matrix between two populations: an operator ODE becomes a population ODE plus a finite matrix. But the matrix acts between *frozen* spans of initialization observables, so the learned middle layer can only rotate within a span fixed at initialization, while $W^{(1)}$ and w move freely. This is a frozen-dictionary closure of the middle layer, and it shows the characteristic limitations of that class: convergence comes only from the spans becoming dense, there is no rate in N , and the guarantee holds on a short horizon near a reference task. The two laws are continuum objects, so the state is not a finite list of scalars. The frozen-dictionary experiments of Section 4 show the same limitation at finite width, and the response memories of Section 3, whose atoms coevolve with the features, are the answer to it.

Details

Population flow. With $\phi = \tanh$, $d = 2$ and inputs $x \in \sqrt{2}S^1$, the population fields are $z^{(1)}(x) = W^{(1)} \cdot x / \sqrt{d}$, $h^{(1)} = \phi(z^{(1)})$, $z^{(2)}(x) = W^{(2)}h^{(1)}(x)$, $h^{(2)} = \phi(z^{(2)})$, $\delta^{(2)}(x) = w\phi'(z^{(2)}(x))$, $\delta^{(1)}(x) = \phi'(z^{(1)}(x))(W^{(2)})^*\delta^{(2)}(x)$ and $f(x) = \mathbb{E}_2[w h^{(2)}(x)]$; with $r = f - y$ and a training law μ ,

$$\dot{W}^{(1)} = -2 \int r \delta^{(1)} \frac{x}{\sqrt{d}} d\mu, \quad \dot{W}^{(2)} = -2 \int r \delta^{(2)} \otimes h^{(1)} d\mu, \quad \dot{w} = -2 \int r h^{(2)} d\mu.$$

Observables. Let $\psi_1 \in \mathbb{R}^{N_1}$ and $\psi_2 \in \mathbb{R}^{N_2}$ be the order- N vectors of initialization observables on the two populations, ridge-normalized by $\psi_\ell \mapsto (\mathbb{E}_\ell[\psi_\ell \psi_\ell^\top] + 2^{-N}I)^{-1/2} \psi_\ell$, so that the projections $g \mapsto \psi_\ell^\top \mathbb{E}_\ell[\psi_\ell g]$ are contractions converging strongly to the identity on the observable spaces. The initialized middle action is recorded in $M_0 = \mathbb{E}_2[\psi_2 (W_0^{(2)} \psi_1)^\top]$; the forward and adjoint actions of $W_0^{(2)}$ on observables are drawn from one correlated Gaussian source, which is what makes the two orientations one operator and its adjoint.

Closure. The state is $\text{Law}(\psi_1, W_0^{(1)}, W^{(1)})$ on population 1, $\text{Law}(\psi_2, w)$ on population 2, and $M \in \mathbb{R}^{N_2 \times N_1}$, initialized with $W^{(1)} = W_0^{(1)}$, $w = 0$ and $M = M_0$. The closure is the population flow above with $W^{(2)}$ replaced by

$$W_N^{(2)} g = \psi_2^\top M \mathbb{E}_1[\psi_1 g], \quad (W_N^{(2)})^* g = \psi_1^\top M^\top \mathbb{E}_2[\psi_2 g],$$

$$\dot{M} = -2 \int r_N \mathbb{E}_2[\psi_2 \delta^{(2)}] \mathbb{E}_1[\psi_1 h^{(1)}]^\top d\mu,$$

so that $z_N^{(2)}(x) = \psi_2^\top M \mathbb{E}_1[\psi_1 h^{(1)}(x)]$ and $f_N(x) = \mathbb{E}_2[w h_N^{(2)}(x)]$; the weights $W^{(1)}$ and w move by the flow equations with r_N and $\delta_N^{(1)}$, which transports the two laws. The represented operator is $W_N^{(2)}$, whose learned part is $M - M_0$ written between observable spaces.

Well-posedness. For fixed N the characteristic equations are locally Lipschitz. The loss obeys $\frac{d}{dt} \mathcal{L}_N = -\left\| \dot{W}^{(1)} \right\|_2^2 - \left\| \dot{M} \right\|_F^2 - \left\| \dot{w} \right\|_2^2$, and $\mathcal{L}_N(0) \leq Y^2$ gives $\|w(t)\|_\infty \leq 2Yt$, $\|M(t) - M_0\|_F \leq 2Y^2 t^2$, $\left\| W_N^{(2)}(t) \right\|_{\text{op}} \leq 2 + 2Y^2 t^2$ and $\|W^{(1)}(t)\|_2 \leq \sqrt{2} + 4Y^2 t^2 + 2Y^4 t^4$, which exclude finite-time escape. The joint laws contain the correlations between observables and current weights, so the saved state suffices to restart.

Convergence. Place closure and population flow on common probability spaces and set

$$e_N = \left\| W_N^{(1)} - W^{(1)} \right\|_2 + \left\| (W_N^{(2)} - W_{N,0}^{(2)}) - (W^{(2)} - W_0^{(2)}) \right\|_{\text{HS}} + \|w_N - w\|_2,$$

where $W_{N,0}^{(2)}$ is the closure's initial operator. The basis error

$$\begin{aligned} \varepsilon_N = & \sup_{t,x} \left\| (W_{N,0}^{(2)} - W_0^{(2)}) h^{(1)}(t, x) \right\|_2 + \sup_{t,x} \left\| (W_{N,0}^{(2)} - W_0^{(2)})^* \delta^{(2)}(t, x) \right\|_2 \\ & + \sup_t \left\| \Pi_2 \dot{W}^{(2)}(t) \Pi_1 - \dot{W}^{(2)}(t) \right\|_{\text{HS}}, \end{aligned}$$

with Π_ℓ the observable projections, tends to zero by strong convergence of the projections, compactness of the population response curves in (t, x) , and density of finite-rank operators. Forward subtraction gives, uniformly in x , $\left\| h_N^{(1)} - h^{(1)} \right\|_2 \leq e_N$ and $\left\| h_N^{(2)} - h^{(2)} \right\|_2 + |f_N - f| \leq C(e_N + \varepsilon_N)$. Backward subtraction multiplies a changed gate by an unbounded reference field; with the exponential tail bound of the reference trajectory, truncating at level R gives

$$e'_N \leq C(1 + R)(e_N + \varepsilon_N) + C e^{-aR}.$$

Choosing $R = 1 + a^{-1} \log(1/v)$ with $v = e_N + \varepsilon_N + \eta$ yields $v' \leq \Lambda_0 v \log(e/v)$, hence

$$e_N(t) + \varepsilon_N + \eta \leq e^{1-\alpha(t)} (\varepsilon_N + \eta)^{\alpha(t)}, \quad \alpha(t) = e^{-\Lambda_0 t}.$$

Letting $\eta \downarrow 0$ and $N \rightarrow \infty$ gives $\sup_{t \leq T} e_N \rightarrow 0$, and the forward bounds give uniform convergence of f_N on $[0, T] \times S^1$. The same induction over observation programs controls every admissible joint observation in L^2 , and the canonical coupling bounds its W_2 distance by the L^2 distance.

Feature learning at ν_* . Write $g_a = W_0^{(1)} \cdot e_a$ for the initial preactivations at the two inputs, $h_a = \phi(g_a)$, $\xi_a = W_0^{(2)} h_a$ and $S = \phi(\xi_1) - \phi(\xi_2)$. With $U_a = S \phi'(\xi_a)$, $P_a = (W_0^{(2)})^* U_a$ and $V_a = \phi'(g_a)^2 P_a$, the population flow satisfies $w(t) = tS + o(t)$, $h^{(1)}(t, e_a) - h_a = \frac{y_a}{2} t^2 V_a + o(t^2)$ and $h^{(2)}(t, e_a) - \phi(\xi_a) = \frac{y_a}{2} t^2 \phi'(\xi_a) (\mathbb{E}[\phi^2(G)] U_a + W_0^{(2)} V_a) + o(t^2)$ for a standard Gaussian G . Adjunction gives $\mathbb{E}[h_a P_a] = y_a \mathbb{E}[\xi_a \phi(\xi_a) \phi'(\xi_a)] \neq 0$, so both hidden displacements are nonzero at order t^2 ; a nondegenerate Gaussian preactivation has positive nonaffinity since $\tanh$ is not affine on any interval; and continuity of the flow in the training law extends both properties to a neighborhood of ν_* .

D Exact compression in a deep linear benchmark

Take one datum $x = 1$, a label y , three linear hidden layers with order-one initialization, mobilities $(n\kappa, \kappa, \kappa, n\kappa)$, and the normalized factors $u = W^{(1)}/\sqrt{n}$, $v = W^{(4)}/\sqrt{n}$, so that $f_n = v^\top W^{(3)} W^{(2)} u$. Packing the four factors into the cyclic block matrix $\mathcal{C}_n$ with blocks $v^\top, u, W^{(2)}, W^{(3)}$ below and above the diagonal gives $f_n = \frac{1}{4} \text{Tr} \mathcal{C}_n^4$, and the feature derivative of $\mathcal{C}_n$ is $(\mathcal{C}_n^\top)^3$.

Theorem D.1 (Exact operator closure). *There are a separable Hilbert space, a bounded initialized source $\mathcal{C}_0$ and a unique global trace-class increment $Q(t)$ with*

$$\mathcal{C} = \mathcal{C}_0 + Q, \quad f = \frac{1}{4} \text{Tr} \mathcal{C}^4, \quad \dot{Q} = -2\kappa(f - y)(\mathcal{C}_0^* + Q^*)^3, \quad Q(0) = 0.$$

The kernel is $K = \text{Tr}[\mathcal{C}^3(\mathcal{C}^)^3]$, the residual obeys $\dot{r} = -2\kappa r K$, and $\|Q(t)\|_1 \leq 2|y|\sqrt{\kappa t}$. For every fixed T , finite gradient flow, and gradient descent for every step sequence tending to zero, converge in probability to $(f, K, \mathcal{L})$ uniformly on $[0, T]$.*

The proof builds $\mathcal{C}_0$ as the limit of initialized word statistics on a colored Fock space, obtains global existence from trace-norm control, passes to the width limit through finite-word Picard iterates, and treats gradient descent as an exact Euler step for Q .

Theorem D.2 (Restricted scalar nonclosure). *In the same model, no encoder formed from a fixed finite list of complete tensor contractions of $(u, W^{(2)}, W^{(3)}, v)$ with connected-graph degree bounded independently of width admits a width-independent polynomial autonomous ODE (or finite-order local polynomial PDE) and a polynomial output readout reproducing the output dynamics on a nonempty open set of states for all sufficiently large widths.*

The $(2j - 1)$ st feature derivative of the output contains the connected contraction

$$u^\top ((W^{(2)})^\top (W^{(3)})^\top W^{(3)} W^{(2)})^j u$$

of degree $4j + 2$. Polynomial operations on a bounded-degree alphabet form only disjoint unions of connected components already present, and nonisomorphic contraction graphs are linearly independent at large width, so this term cannot be synthesized. The theorem does not exclude arbitrary encoders, unrestricted fields, or closures tailored to a single initialized trajectory.

E Accuracy and resource comparisons: derivations

We compare approximations to the same population predictor f_∞ of (2), assumed to exist on $[0, T]$, with the same data and initialization law, including the readout. Matching initial predictions alone does not match the initial tangent kernel. Throughout, constants can depend on m, L, T , the data, initialization and test law; displayed factors in these variables do not exhaust that dependence. Population error estimates below are stated hypotheses, whereas the fixed-width response-memory results are Theorems 3.1 and 3.2. We suppress the T subscripts on error constants.

Size at accuracy. The main text compares descriptions by the size of their moving state and the rate at which their error falls in their order. The quantity optimized below is the following.

Definition E.1 (Size at accuracy). For a specified family of representations, a horizon T and an accuracy ε , let $M(\varepsilon; T)$ be the least moving state obtained by choosing its resolution parameters to satisfy a stated sufficient error bound on $[0, T]$. This is a certified representation cost, not a lower bound on all algorithms.

The error is measured in the common metric (38) below, against a population predictor f_∞ on $[0, T]$, assumed to exist.

Conditional population hypotheses. The quantitative population comparison uses the following assumptions where applicable; the derivations below specify their role.

- (A1) A matched width approximation in the common metric (38): $\mathcal{E}_Q(f_n; T) \leq A/\sqrt{n}$.
- (A2) A same-initialization tracking estimate $(\mathbb{E} \sup_{t \leq T} \|\hat{f}_{n,q}(t) - f_n(t)\|_{L^2(Q)}^2)^{1/2} \leq B_\omega/q^\omega$, for $\omega = 1, 2$, with constants and sufficient-order thresholds uniform in width. Together with (A1), this gives $\mathcal{E}_Q(\hat{f}_{n,q}; T) \leq A/\sqrt{n} + B_\omega/q^\omega$.
- (A3) Stable numerical discretization, sampling and self-consistency for the specified limiting-response solver, giving $A_D/\sqrt{S} + B_D(T/N_t)^\sigma + \eta$.

(A4) Matched population tangent kernels satisfying $B_j \leq UV^{j-2}(j-2)!$ and $\chi = 2RVT < 1$ as defined below, together with access to their initial coefficients and mixed test-input evaluators.

These are hypotheses for the stated common target, not additional population theorems proved here. In particular, numerical solver convergence and width convergence are distinct approximation problems.

E.1 Test error, allocations and arithmetic

For a test probability measure Q and a candidate predictor g , define

$$E_Q(g; T) = \sup_{t \leq T} \|g(t) - f_\infty(t)\|_{L^2(Q)}, \quad \mathcal{E}_Q(g; T) = (\mathbb{E} E_Q(g; T)^2)^{1/2}. \quad (38)$$

The reverse triangle inequality bounds the difference between the test RMSEs of g and f_∞ , against any $y_* \in L^2(Q)$, by $E_Q(g; T)$. This measures approximation, without assuming that f_∞ generalizes well. Moreover, pointwise estimates $\mathbb{E} \sup_{t \leq T} |f_n(t, x) - f_\infty(t, x)|^2 \leq C_T(x)^2/n$ with $C_T \in L^2(Q)$ give $\mathcal{E}_Q(f_n; T)^2 \leq n^{-1} \int C_T^2 dQ$ by moving the time supremum inside the integral and applying Tonelli. An input supremum would require additional spatial control.

For $\phi = \tanh$, set $a_\ell = \|h^{(\ell)} - h'^{(\ell)}\|_2 / \sqrt{n}$ and assume $\|x\| / \sqrt{d} \leq X$. Boundedness and Lipschitz continuity of $\tanh$ give

$$a_1 \leq X \frac{\|W^{(1)} - W'^{(1)}\|_F}{\sqrt{n}}, \quad a_\ell \leq \|W^{(\ell)} - W'^{(\ell)}\|_{\text{op}} + \|W'^{(\ell)}\|_{\text{op}} a_{\ell-1},$$

$$|f_n - f'_n| \leq \frac{\|w - w'\|_2}{\sqrt{n}} + \frac{\|w'\|_2}{\sqrt{n}} a_L.$$

Induction transfers parameter control to every such test input when operator and normalized readout norms are bounded. Coupling dense and compressed networks through their initialization, the triangle inequality and Minkowski give

$$\mathcal{E}_Q(\hat{f}_{n,q}; T) \leq \mathcal{E}_Q(f_n; T) + \left(\mathbb{E} \sup_{t \leq T} \|\hat{f}_{n,q}(t) - f_n(t)\|_{L^2(Q)}^2 \right)^{1/2}. \quad (39)$$

Thus the conditional model $An^{-1/2} + B_\omega q^{-\omega}$, $\omega = 1, 2$, needs both a matched root-width estimate and closure constants and sufficient-order thresholds uniform in width, with appropriate moments. The perturbative fluctuation calculation of Bordelon and Pehlevan (2023) motivates the first estimate but is not a general remainder theorem for this model.

Moving memory counts persistent updated scalars; fixed matrices, coefficient evaluators and workspace are reported separately. The leading response-memory count is $2(L-1)mnq$. With $u = A/\sqrt{n}$, $v = B_\omega/q^\omega$ and $u + v = \varepsilon$, its minimization reduces to minimizing $u^{-2}(\varepsilon - u)^{-1/\omega}$. The logarithm has derivative $-2/u + 1/[\omega(\varepsilon - u)]$ and positive second derivative, and diverges at both endpoints. Hence the unique continuous optimum is

$$u_* = \frac{2\omega\varepsilon}{2\omega+1}, \quad v_* = \frac{\varepsilon}{2\omega+1}, \quad n_* = \frac{(2\omega+1)^2 A^2}{4\omega^2 \varepsilon^2}, \quad q_* = \left(\frac{(2\omega+1)B_\omega}{\varepsilon} \right)^{1/\omega}. \quad (40)$$

For $\omega = 1$, $(n_*, q_*) = (9A^2/(4\varepsilon^2), 3B_1/\varepsilon)$ and $M_* \sim \frac{27}{2}(L-1)mA^2B_1\varepsilon^{-3}$. For $\omega = 2$, $(n_*, q_*) = (25A^2/(16\varepsilon^2), \sqrt{5B_2}/\varepsilon)$ and $M_* \sim \frac{25\sqrt{5}}{8}(L-1)mA^2\sqrt{B_2}\varepsilon^{-5/2}$. Upward rounding is feasible. Outer weights, $(d+1)n$, and the joint Gram matrix, $O(q^2)$, perturb the finite-tolerance optimum but are lower order. Dense training instead takes $n = \lceil (A/\varepsilon)^2 \rceil$ and $M_* \sim (L-1)A^4\varepsilon^{-4}$. These optimize sufficient bounds in specified representations, not actual errors or all possible algorithms.

Relative saving with the original dense initialization. For the quadratic rate, the ratio of these optimized certificates is

$$\frac{M_{\text{memory}}^*(\varepsilon)}{M_{\text{dense}}^*(\varepsilon)} \sim \frac{25\sqrt{5}}{8} \frac{m\sqrt{B_2}}{A^2} \varepsilon^{3/2}. \quad (41)$$

This proves Corollary 3.5’s asymptotically strict learned-state saving under its two error hypotheses. The optimized closure uses only 25/16 times the dense certificate’s width, to leading order; its memory order grows as the fourth root of width. For a first-order tracking estimate the corresponding ratio is $\frac{27}{2}mB_1A^{-2}\varepsilon$. Neither ratio compares two upper bounds as though they were lower bounds: the denominator is the actual state count of the dense representation sized by (A1), and the numerator is the count sized by (A1)–(A2). A lower bound on the width truly necessary for dense prediction would be a separate result.

The same calculation works with a near-quadratic tracking estimate $Bq^{-2}\exp(K\sqrt{\log(e+q)})$, for fixed $B > 0$, $K \geq 0$. For any fixed $D > K/(2\sqrt{2})$, take, for sufficiently small ε ,

$$n = \left\lceil \frac{25A^2}{16\varepsilon^2} \right\rceil, \quad q = \left\lceil \varepsilon^{-1/2} e^{D\sqrt{\log(1/\varepsilon)}} \right\rceil.$$

Writing $s = \log(1/\varepsilon)$, the history error is bounded by

$$B\varepsilon \exp\left[(-2D + K/\sqrt{2})\sqrt{s} + O(1)\right] \leq \varepsilon/5.$$

Consequently the achievable moving count and relative count satisfy

$$\begin{aligned} M_{\text{memory}} &= O\left(Lm\varepsilon^{-5/2}e^{D\sqrt{\log(1/\varepsilon)}}\right), \\ \frac{M_{\text{memory}}}{M_{\text{dense}}} &= O\left(m\varepsilon^{3/2}e^{D\sqrt{\log(1/\varepsilon)}}\right) \rightarrow 0. \end{aligned}$$

The first expression has exponent $5/2 + o(1)$ in $1/\varepsilon$, approaching $5/2$ from above; it need not be $O(\varepsilon^{-5/2})$ itself.

These implications also hold for the all-time metric obtained by replacing $t \leq T$ with $t \geq 0$, provided (A1)–(A2), including their joint admissible width/order range, hold in that metric with finite constants independent of the horizon. If the predictors converge at the end of training, their limits inherit the same $L^2(Q)$ prediction bound. For any $y_* \in L^2(Q)$, the reverse triangle inequality transfers it to the discrepancy between their test RMSEs. Separate limits at fixed q followed by $q \rightarrow \infty$ do not justify the simultaneous choice $q \asymp n^{1/4}$; the joint estimate in (A2) is essential. This argument retains exactly the canonical dense W_0 and does not invoke a block-initialization approximation.

For completeness, full-batch arithmetic per right-hand-side evaluation is

$$\begin{aligned} C_{\text{dense}} &= O(Lmn^2 + mnd), \\ C_{\text{old}} &= O(Lmn^2 + Lm^2nq + Lmnq + mnd), \\ C_{\text{joint}} &= O(Lmn^2 + Lm^2nq + Lmnq^2 + q^3 + mnd). \end{aligned} \quad (42)$$

The terms respectively account for dense initialized actions, learned factors, moment updates or Gram solves, and Gram factorization. Directional passes for the joint clock add a constant number of network passes. Both closures retain $(L-1)n^2$ fixed entries; activation workspace is $O(Lmn)$, and Gram conditioning can affect numerical step counts. Consequently neither their total-storage nor their arithmetic exponent improves automatically.

E.2 A matched tangent hierarchy

With $\mathcal{D}_n = \text{diag}(nI, I, \dots, I, nI)$, define recursively

$$\begin{aligned} K^{(2)}(x, z) &= \nabla_{\theta} f(x)^{\top} \mathcal{D}_n \nabla_{\theta} f(z), \\ K^{(j+1)}(x_1, \dots, x_{j+1}) &= (\nabla_{\theta} K^{(j)}(x_1, \dots, x_j))^{\top} \mathcal{D}_n \nabla_{\theta} f(x_{j+1}). \end{aligned}$$

The chain rule gives the exact, mobility-matched hierarchy

$$\dot{f}(x) = -\frac{2}{m} \sum_a r_a K^{(2)}(x, x_a), \quad \dot{K}^{(j)} = -\frac{2}{m} \sum_a r_a K^{(j+1)}(\dots, x_a). \quad (43)$$

Truncation through order $q \geq 2$ retains exact initial coefficients and freezes $K^{(q)}$. A population version additionally requires limiting coefficients and a valid limiting hierarchy.

Feedback bound. Suppose $|K^{(j)}| \leq B_j$ along the exact flow, uniformly when the first argument is a training or test input and the others are training inputs. Let $R = 1 + \sup_{t,a} |r_a(t)|$ and stop the truncated system at predictor error one. Its residuals are then bounded by R . For supremum predictor error e_f and order- j kernel error e_j , subtraction, splitting each product using the exact kernel and truncated residual, yields

$$\begin{aligned} e_q(t) &\leq 2RB_{q+1}t, \\ e_j(t) &\leq \int_0^t (2Re_{j+1} + 2B_{j+1}e_f) ds \quad (2 \leq j < q), \\ e_f(t) &\leq \int_0^t (2Re_2 + 2B_2e_f) ds. \end{aligned} \quad (44)$$

Successive substitution integrates the top forcing q times. Each feedback term has convolution kernel $2B_{k+2}(2R(t-s))^k/k!$, $0 \leq k \leq q-2$. Bounding these kernels at T and applying Gronwall gives

$$\sup_{t \leq T} e_f(t) \leq \eta_q(T) := B_{q+1} \frac{(2RT)^q}{q!} \exp(\Lambda_{q,T} T), \quad \Lambda_{q,T} = 2 \sum_{k=0}^{q-2} B_{k+2} \frac{(2RT)^k}{k!}. \quad (45)$$

If $\eta_q(T) < 1$, first exit is impossible. The triangular integral equations then bound all training tensors, preventing finite-time escape and extending the solution through T . The test bound implies $E_Q \leq \eta_q$ for Q supported on the controlled test set. For random coefficients, the same probabilistic convention and coefficient errors must also be controlled.

One sufficient regime is $B_j \leq UV^{j-2}(j-2)!$ with $U, V > 0$ and $\chi = 2RVT < 1$. Then $\Lambda_{q,T} \leq 2U/(1-\chi)$ and $\eta_q \leq D\chi^q/q$, where $D = (U/V)e^{2UT/(1-\chi)}$. For $0 < \varepsilon < 1$, put $\lambda_{\chi} = \log(1/\chi)$. The smallest certified integer order is

$$q_{\varepsilon} = \max \left\{ 2, \left\lceil \frac{W(\lambda_{\chi} D / \varepsilon)}{\lambda_{\chi}} \right\rceil \right\}, \quad M_{\text{NTH}} = \sum_{j=1}^{q_{\varepsilon}-1} m^j = \frac{m^{q_{\varepsilon}} - m}{m - 1} \quad (m \geq 2). \quad (46)$$

Here $W(z)e^{W(z)} = z$: the error inequality is equivalent to $(\lambda_{\chi} q)e^{\lambda_{\chi} q} \geq \lambda_{\chi} D / \varepsilon$. The top tensor has $m^{q_{\varepsilon}}$ fixed entries, and direct work is $O(m^{q_{\varepsilon}})$ per evaluation. Depth is already combined into these kernels. For fixed $m \geq 2$ and χ , the moving count is $\asymp [D/(\varepsilon \log(D/\varepsilon))]^{\log m / \lambda_{\chi}}$. Tanh analyticity alone does not supply these uniform kernel bounds; $\chi \geq 1$ makes this certificate inconclusive. Population use requires U, V, R independent of the approximation resources. Finite-width initial coefficients also require a width-error allowance; with population coefficients, their construction and approximation are fixed-resource costs.

Scope of the published estimate. After exchanging their width/sample symbols, Huang and Yau (2020, Theorem 2.6) give the training RMS bound

$$\begin{aligned} \frac{\|f(t) - \tilde{f}(t)\|_2}{\sqrt{m}} &\lesssim \frac{(1+t)t^{q-1}}{n^{q/2}} \min\{t, m/\lambda\}, \\ t &\leq \min \left\{ \frac{c\sqrt{\lambda n/m}}{(\log n)^C}, \frac{n^{q/[2(q+1)]}}{(\log n)^{C'}} \right\}. \end{aligned} \quad (47)$$

Time uses the source loss convention; a factor-two time rescaling changes constants, not these powers. This uses their high-probability initialization setting and parameterization, even q , bounded activation derivatives through $2q + 1$, nondegenerate bounded inputs with uniformly independent subsets through size $2q + 1$, and an initial kernel eigenvalue exceeding λ . Three vectors in two dimensions violate the subset assumption. Their small higher-kernel estimates do not establish (45)’s population hypotheses for our feature-learning flow; their training RMS estimate is not a whole-test-function estimate.

Whole-function evaluation and time analyticity. Set $u_a = -2\tilde{r}_a/m$, $I_\emptyset = 1$, and $\dot{I}_{a_1 \dots a_k} = u_{a_1} I_{a_2 \dots a_k}$ with zero initial values for nonempty words. Repeated integration of the triangular system gives

$$\tilde{f}_q(t, x) = f_0(x) + \sum_{k=1}^{q-1} \sum_{a_1, \dots, a_k} K_0^{(k+1)}(x, x_{a_1}, \dots, x_{a_k}) I_{a_1 \dots a_k}(t). \quad (48)$$

This uses the same number of moving coefficients as the direct hierarchy; one query needs that many contractions *plus* the cost of the fixed mixed-kernel evaluators. When $m = 1$, $I_k = s^k/k!$ and $\dot{s} = u$, so one moving scalar suffices and order dependence remains in fixed coefficients and evaluation. These counts are therefore representation costs, not lower bounds.

Truncation is not a time-Taylor polynomial. For the scalar model $f(\theta) = \theta^2$, loss f^2 , target zero and unit mobility, $f_0 > 0$ gives $\dot{\theta} = -4\theta^3$, $f(t) = f_0/(1 + 8f_0t)$ and $K^{(j)} = 4^{j-1}f$. Its time-Taylor radius is $1/(8f_0)$, but (48) becomes $\tilde{f}_q = f_0 \sum_{k=0}^{q-1} (4s)^k/k!$, $\dot{s} = -2\tilde{f}_q$. The exact scalar field replaces the sum by e^{4s} , with solution $s = -\frac{1}{4} \log(1 + 8f_0t)$. On a compact neighborhood of this path for any finite T , the polynomial fields and their derivatives converge uniformly. Subtracting the ODEs gives a vanishing forcing and a uniform Lipschitz coefficient; Gronwall and first exit give convergence on $[0, T]$. Thus a finite time-Taylor radius is no universal barrier for NTH.

E.3 A specified streamed DMFT solver

The sampled self-consistency algorithm of Bordelon and Pehlevan (2022, Appendix B, Algorithm 1) stores layerwise covariance and response kernels indexed by input and time pairs. Each sweep updates predictions, factors Gaussian covariances, samples paths, solves their causal equations and response Jacobians, and averages products and derivatives. Matching its feature-learning scale to ours requires $\gamma = \sqrt{n}$, internal coordinates $U^{(\ell)} = \sqrt{n}W^{(\ell)}$, and the same readout law and loss normalization. A fixed-program Tensor Programs limit (Yang and Hu, 2021, Algorithm 1, Theorem 7.4) does not itself prescribe this numerical solver or a uniform numerical error estimate.

For N_t time steps, S paths and I sweeps, a dense covariance factor costs $O((mN_t)^3)$ per layer. Dense triangular solves for a full samplewise response Jacobian cost at most the same per path. Streaming one path and Jacobian while accumulating kernel averages therefore permits

$$M_D = O(Lm^2N_t^2), \quad C_{D,\text{total}} = O(IL(S+1)m^3N_t^3), \quad C_{D,\text{step}} = O(IL(S+1)m^3N_t^2). \quad (49)$$

The last cost is amortized; memory includes kernel and Jacobian workspace. These are implementable upper bounds, not framework-wide lower bounds. Passive test prediction requires mixed kernels and test paths as well.

To justify an error model, suppose the discrete self-consistency map $\mathcal{F}$ is a κ -contraction, $\kappa < 1$, on a common domain, in a norm controlling test outputs, has fixed point z_* , and the sampled map $\hat{\mathcal{F}}$ differs uniformly by at most ζ . For a computed point $\hat{z}$ with residual $\|\hat{z} - \hat{\mathcal{F}}(\hat{z})\| \leq \eta_0$, the triangle inequality gives $\|\hat{z} - z_*\| \leq \eta_0 + \zeta + \kappa \|\hat{z} - z_*\|$. A Lipschitz output map, uniform RMS sampling control and a stable order- σ time approximation would thus yield

$$\mathcal{E}_Q \leq A_D S^{-1/2} + B_D (T/N_t)^\sigma + \eta, \quad \eta \leq C\eta_0/(1 - \kappa). \quad (50)$$

The constants include stability amplification. The cited papers do not establish these hypotheses uniformly as the grid grows for this model. Setting $S = n$ matches a sampling exponent, not the covariance of finite-network fluctuations (Bordelon and Pehlevan, 2023).

Since streaming makes leading memory independent of S , its continuous memory infimum has $N_{t,\text{inf}} = T(B_D/\varepsilon)^{1/\sigma}$ and $M_{\text{inf}} \asymp Lm^2 T^2 B_D^{2/\sigma} \varepsilon^{-2/\sigma}$. A finite feasible allocation reserves $\beta\varepsilon$, $0 < \beta < 1$, for sampling and iteration: take $N_t = \lceil T[B_D/((1 - \beta)\varepsilon)]^{1/\sigma} \rceil$, $S \geq (2A_D/(\beta\varepsilon))^2$, and $\eta \leq \beta\varepsilon/2$. Approaching the infimum increases work. Optimizing numerical order requires additional regularity, constants and stage costs; there is no intrinsic DMFT accuracy exponent here.

A distinct finite-width baseline stores every Euler update exactly:

$$W_k^{(\ell)} = W_0^{(\ell)} - \frac{2\Delta}{nm} \sum_{j < k, a} r_{a,j} \delta_{a,j}^{(\ell)} (h_{a,j}^{(\ell-1)})^\top.$$

Its moving memory is $O(LmnN_t)$, fixed memory $O(Ln^2)$ and late-step work $O(Lmn^2 + Lm^2 n N_t)$. A fixed-stage order- σ method has the same storage order; with error $A/\sqrt{n} + D_h(T/N_t)^\sigma$, optimizing as above gives $M_* \asymp LmA^2 T D_h^{1/\sigma} \varepsilon^{-(2+1/\sigma)}$. This stores finite-network updates and is not Tensor Programs sampling.

The optimized costs together. Suppressing constant factors at fixed task and horizon gives

Representation	Leading moving state at error ε	Conditions
Dense	$LA^4 \varepsilon^{-4}$	(A1)
Learning-speed memory	$LmA^2 B_1 \varepsilon^{-3}$	(A1), (A2)
Joint response memory	$LmA^2 \sqrt{B_2} \varepsilon^{-5/2}$	(A1), (A2)
NTH, $m \geq 2$	$\sum_{j=1}^{q_\varepsilon-1} m^j$	(A4)
Streamed DMFT	$Lm^2 T^2 B_D^{2/\sigma} \varepsilon^{-2/\sigma}$	(A3)
Stored network history	$LmA^2 T D_h^{1/\sigma} \varepsilon^{-(2+1/\sigma)}$	Matched width/time rates

Response memory separates the cost of sampling width from the cost of representing history: an order rate $q^{-\omega}$ combines with root-width sampling to give $\varepsilon^{-2-1/\omega}$ in this representation. The hierarchy pays instead in sample tuples; the streamed solver pays in two-time kernels and sampling work. Explicit powers of T are properties of these formulas, not a measure of total horizon dependence: every row's constants and admissible resolutions can depend on T, m, L and the task. The comparison therefore supplies an organizing principle rather than a universal ordering of methods.

E.4 A sharper estimate for the original joint clock

Keep exactly $\dot{\tau} = \rho + \|\dot{\Psi}\|_2$, with its matching prefix. Define

$$M_T = 1 + \int_0^T \rho \, dt, \quad V_T = \frac{1}{\sqrt{nm}} \int_0^T \|\dot{\Psi}\|_2 \, dt, \quad \tau_T = M_T + \sqrt{nm} V_T.$$

The insertion measure has mass M_T , while the normalized history $F(\xi) = \Psi(\xi)/\sqrt{nm}$ is $1/\sqrt{nm}$ -Lipschitz.

We need a dimension-independent polynomial approximation fact. If a Hilbert-valued F is L -Lipschitz on $[0, \tau]$, then some polynomial of degree below q has uniform error at most $C_J L \tau / q$. To prove it, extend $G(v) = F(\tau(1 + \cos v)/2)$ evenly and periodically and convolve with the normalized positive trigonometric polynomial $J_N(v) = Z_N^{-1} [\sin(Nv/2)/\sin(v/2)]^4$. On $[-\pi, \pi]$ the ratio is bounded by $\min(N, \pi/|v|)$ and is at least cN near zero. Splitting its integrals at $1/N$ gives $Z_N \asymp N^3$ and $\int |v| J_N(v) \, dv \leq C/N$. The Hilbert norm triangle inequality bounds convolution error by $CL\tau/N$. The convolution is even, hence a polynomial in $\cos v$ of degree at most $2N - 2$. Take $N = \lfloor (q+1)/2 \rfloor$ for $q \geq 2$; a constant polynomial handles $q = 1$.

The weighted projection is no worse than this uniform approximation in $L^2(\mu_T)$. Using the projection energies and product defect of (20) therefore gives

$$\int_0^T \sum_{\ell=2}^L \|E_\ell\|_F \, dt \leq \frac{1}{nm} \sum_{\ell,a} (D_{h,\ell,a} + D_{b,\ell,a}) \leq \frac{C_J^2 M_T}{q^2} \left(\frac{M_T}{\sqrt{nm}} + V_T \right)^2. \quad (51)$$

The first inequality is Cauchy–Schwarz in time followed by $2\sqrt{ab} \leq a + b$; the second uses the actual insertion mass rather than Lebesgue mass τ_T . No clock or algorithm has changed. Width-uniform history mass, normalized variation, feedback stability and continuation thresholds would turn this into a uniform prediction estimate. Establishing those controls, and the matched width rate, remains the population gap.

F Supplementary fitted functions on the sphere

Figure 5 adds a three-dimensional-input, four-hidden-layer example. The sphere RMS discrepancies from dense are 0.02193, 0.01280, and 0.00454 for $q = 1, 2, 3$. Dense and the three closures reach training RMS ≤ 0.01 at times 60, 75.5, 63.25, and 60, respectively. This is supplementary fitted-function evidence outside the smooth-activation theorems: ReLU is nonsmooth, and the endpoints are individually fitted. Rendering the saved predictions does not independently reproduce training.

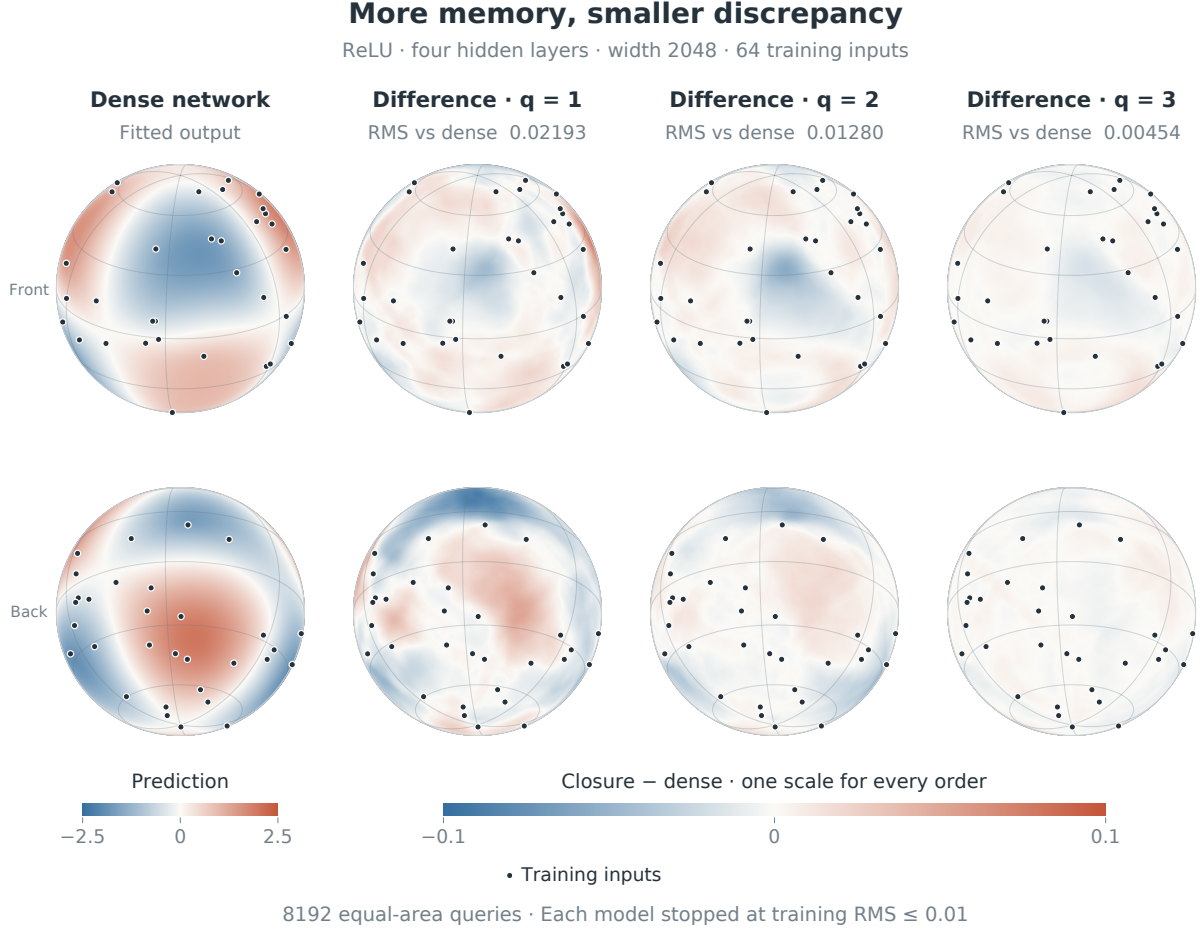


Figure 5: Saved fitted functions on the unit sphere for the target $\sqrt{105}x_1x_2x_3$, with four hidden ReLU layers of width 2048 and 64 training inputs. The first column shows the dense output; the remaining columns show signed closure-minus-dense differences for $q = 1, 2, 3$. Rows show complementary hemispheres, and black dots mark training inputs. Output colors use $[-2.5, 2.5]$; every error panel uses $[-0.1, 0.1]$. Surface colors interpolate saved queries on a triangular mesh; RMS values use the original 8192 equal-area query values before interpolation. Models are compared at their individual training-RMS ≤ 0.01 endpoints, using one network seed and float32 Euler steps of $1/128$. The order trend is specific to this one-seed comparison. These are fitted-function discrepancies, without a common-time trajectory comparison or a continuous-gradient-flow refinement certificate.

Table 2: Whole-circle RMS difference between response memory and dense training at individually fitted endpoints. Left: two hidden tanh layers, width 2048 (Figure 6). Right: three hidden tanh layers, width 4096 (Figure 2).

Task	two layers			three layers		
	$q = 1$	$q = 3$	$q = 7$	$q = 1$	$q = 2$	$q = 3$
Two outliers, alternating	0.236	0.0730	0.00466	0.108	0.0238	0.0315
Quadrant, alternating	0.341	0.0456	0.00155	0.275	0.0284	0.0585
Quadrant, paired	0.0193	0.00282	0.00012	0.0122	0.00371	0.00121
Quadrant, center/edges	0.0456	0.00488	0.00131	0.0790	0.0257	0.00725
Equally spaced, mixed	0.0124	0.00053	4×10^{-6}	0.0150	0.00241	0.00028

G Supplementary two-hidden-layer circle gallery

Figure 6 shows the full two-hidden-layer task gallery, and Table 2 collects the whole-circle RMS discrepancies of both depths (the three-hidden-layer gallery is Figure 2 in the main text).

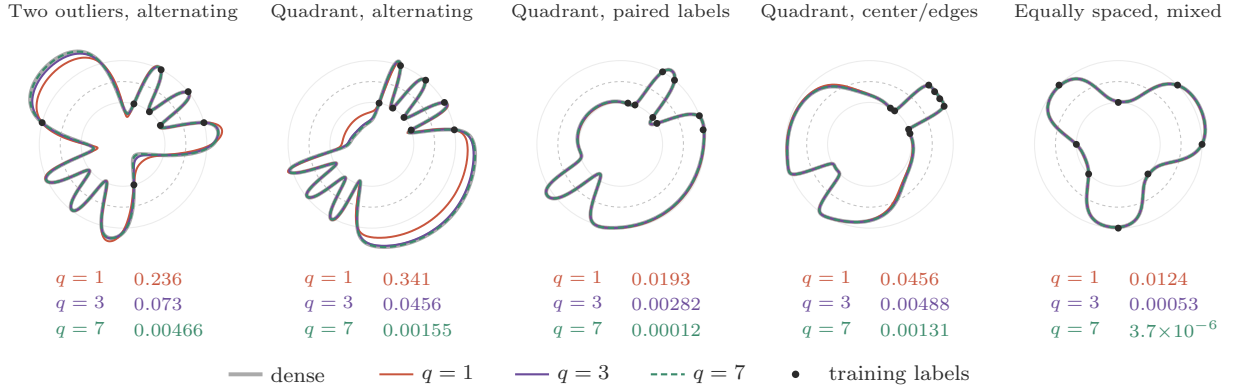


Figure 6: Fitted circle predictions for two hidden tanh layers, width 2048, and response-memory orders $q = 1, 3, 7$, using the dense references in Table 2. Models are evaluated at their individual training-MSE 10^{-3} endpoints. Angle is input direction and radius is $3 + f(\theta)$; rings, training labels and RMS values follow Figure 2. RMS is measured on 8192 uniform circle points.

H Supplementary experimental figures

Figure 7 sketches the paired memory and the product form of its defect; Figure 8 shows what Legendre moments retain of recorded response histories; Figure 9 shows the RMS discrepancies behind Figure 1 at every shared time; Figure 10 shows the three-hidden-layer order trends of Table 2 next to their numerical sensitivity and the movement of the hidden features; Figure 11 shows the MNIST predictions and residuals summarized in Section 4.

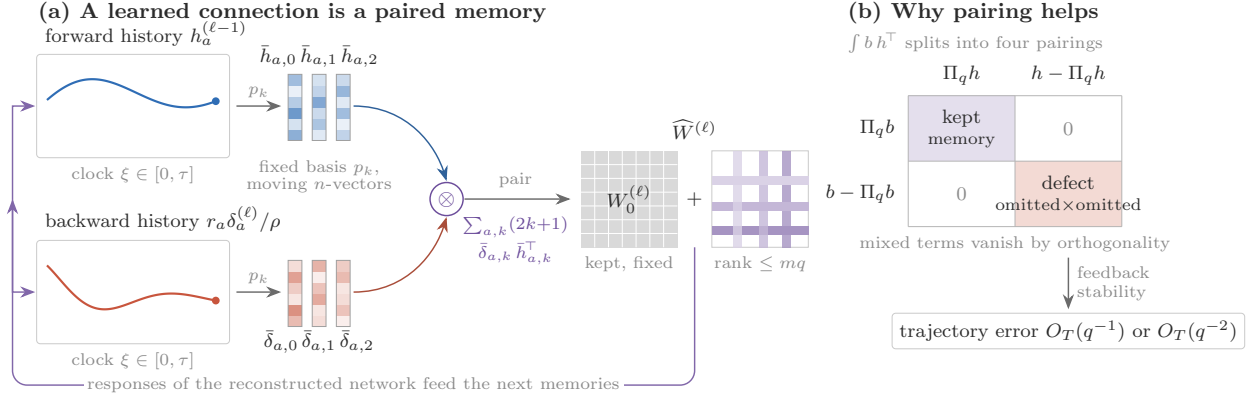


Figure 7: A learned connection is a paired memory. (a) For each hidden link and training sample, the forward response $h_a^{(\ell-1)}$ and the residual-weighted backward response $b_a = r_a \delta_a^{(\ell)} / \rho$ are recorded in the learning clock and projected onto the first q shifted Legendre polynomials ($q = 3$ shown). The polynomial coordinates are fixed; the moment vectors $\bar{h}_{a,k}, \bar{\delta}_{a,k} \in \mathbb{R}^n$ keep moving. Pairing them gives a correction of rank at most mq , added to the fixed initialization $W_0^{(\ell)}$, and the responses of the reconstructed network feed the next memories. (b) Splitting each history into its projection Π_q and remainder, the exact interaction $\int b h^\top$ is a sum of four pairings; the mixed ones vanish by orthogonality, so the memory misses only the product of the two omitted parts.

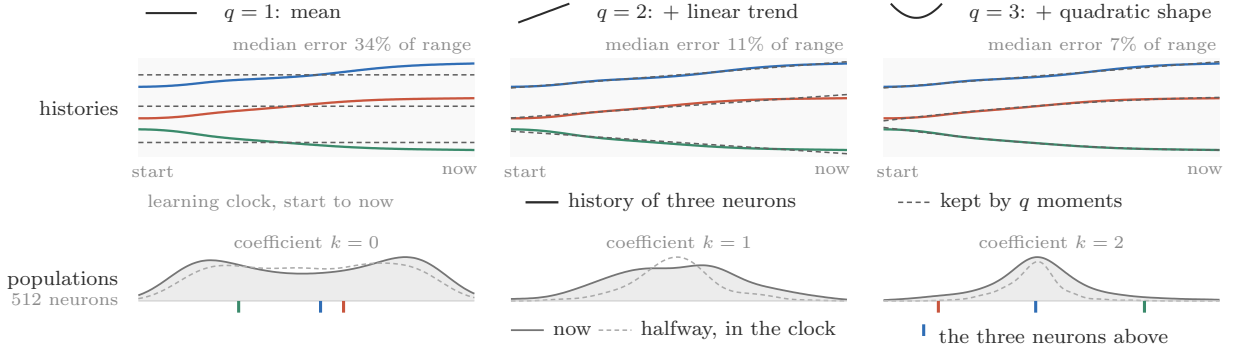


Figure 8: What Legendre moments retain of a history. A network with two hidden tanh layers of width 512 is trained by gradient descent on eight circle points, and the forward responses h of one training input are recorded. Top: three neurons over the whole of training in the learning-speed clock, with their projections onto $q = 1, 2, 3$ Legendre modes (mean, linear trend, quadratic shape). These are projections fitted afterwards to the recorded dense histories, not trajectories of the closure. Above each panel, the median over all 512 neurons of the RMS projection error divided by the range of that neuron's history; for the backward histories $r\delta/\rho$ the same numbers are 26%, 12% and 10%. Bottom: the plotted quantities are the normalized Legendre coefficients $(2k+1)\bar{h}_k/\tau$ of the fitted projection, not the stored moments $\bar{h}_k$; across neurons each coefficient is a population, shown for the history up to the midpoint of the learning clock and for the whole history, with the three neurons above marked.

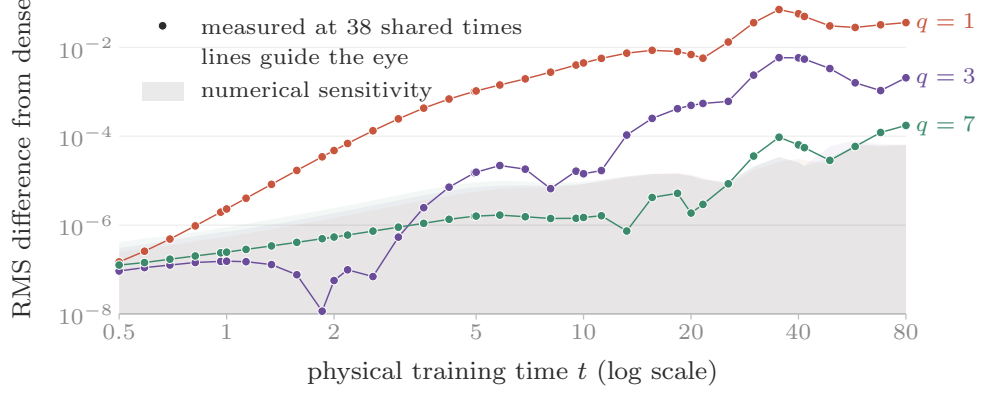


Figure 9: Discrepancy from dense training at common physical times. The paired-label task of Figure 1: RMS difference between response memory and the finer dense replay on 2048 circle queries at the 38 positive shared times, on a logarithmic time axis; the common initialization at $t = 0$ is omitted. Dots are measured; thin lines guide the eye. Light shading extends from the axis floor to each order’s coarse/fine sensitivity scale, obtained by summing the dense and closure prediction changes between the two integration tolerances; it is a numerical diagnostic, not a confidence interval or a certified error bound, and the smallest discrepancies of $q = 3$ and $q = 7$ lie at that scale.

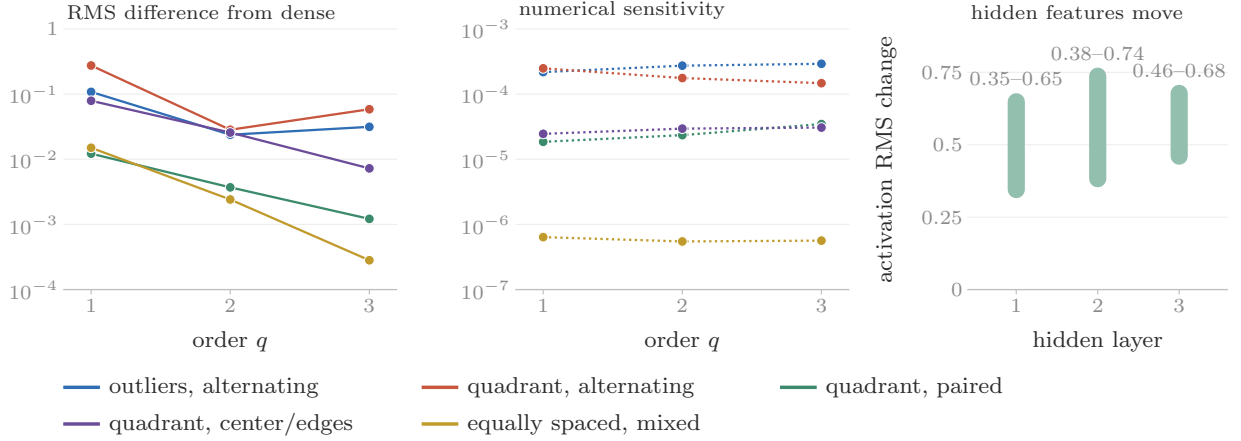


Figure 10: Measured order trends and substantial feature movement. Three-hidden-layer circle experiments, width 4096, at individually fitted MSE 0.001 endpoints. Left: discrepancies on 8192 angles. Middle: sums of dense and closure coarse/fine prediction RMS changes, shown on a separate scale; these are numerical diagnostics, not confidence intervals or certified bounds. Right: ranges of absolute training-activation RMS changes from initialization across the 20 finest runs. Neither the endpoint comparisons nor the three tested orders establish a temporal tracking exponent.

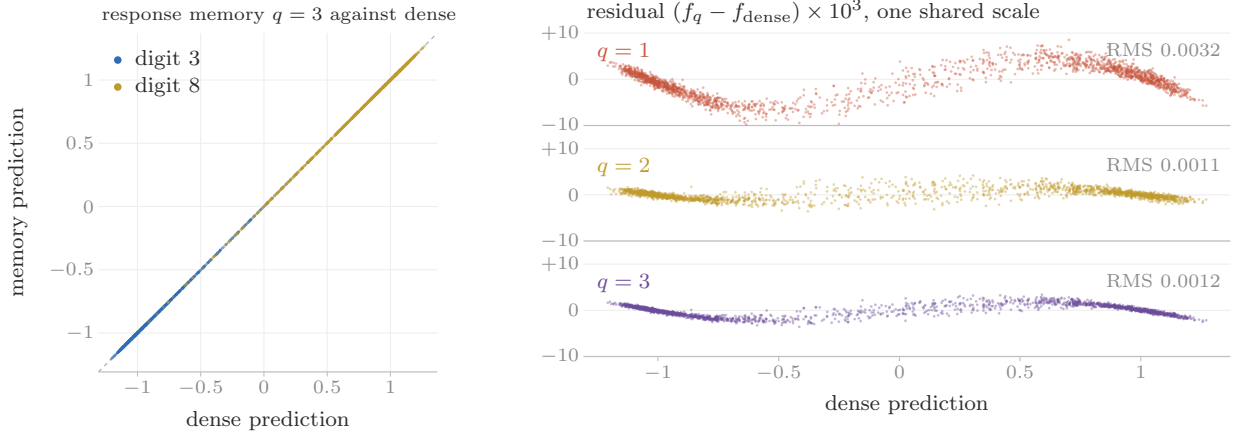


Figure 11: Agreement and its residual structure on MNIST. Digits 3 versus 8, 100 training images, two hidden tanh layers of width 4096. Left: memory $q = 3$ versus dense predictions on all 1984 held-out images. Right: signed differences $f_q - f_{\text{dense}}$, displayed in units of 10^{-3} with one vertical scale for all orders. RMS differences are 0.003248, 0.001084 and 0.001199. Each model is evaluated at its own training-MSE 0.001 endpoint. These are function discrepancies, not label RMSE.

I Frozen-kernel and same-rank controls

The frozen control uses the empirical tangent kernel of the same initialized finite network, including the learning-rate metric. If $J_0(x)$ is its row parameter Jacobian and D has block mobilities $(n, 1, n)$, then $K_0(x, x') = J_0(x)DJ_0(x')^\top$. For the training list X and unhalved mean squared loss, $\dot{f}(X, t) = -(2/m)K_0(X, X)(f(X, t) - y)$. Writing $K_0(X, X) = V \text{diag}(\lambda_j)V^\top$, the test prediction is

$$f_{\text{NTK}}(x, t) = f_0(x) + K_0(x, X)V \text{diag}\left(\frac{1 - e^{-2\lambda_j t/m}}{\lambda_j}\right)V^\top(y - f_0(X)).$$

We evaluate this expression directly and locate the training-MSE 0.001 crossing by bisection, without ridge regularization or discarded eigenmodes. For the equally spaced odd task, the four-representative antipodal quotient removes exact redundancy and preserves the original eight-input mean loss. Kernel blocks are checked against automatic differentiation, and the spectral solution against a matrix exponential and training cross-kernel predictions. The center/edges kernel is ill-conditioned (condition number 2.29×10^9); the two training-prediction formulas agree within 2.1×10^{-8} .

Figures 12 and 13 retain the compact and detailed comparisons with direct factor training. Figure 14 adds the frozen kernel on all five tasks, keeping $q = 3$ and both factor seeds throughout. Memory is closest to the dense endpoint in every row. These are fitted-function comparisons at individual stopping times; the kernel is that of this feature-learning parameterization, rather than a separately scaled infinite-width lazy model. All source predictions, kernel calculations, and reproduction commands are retained with the figure scripts.

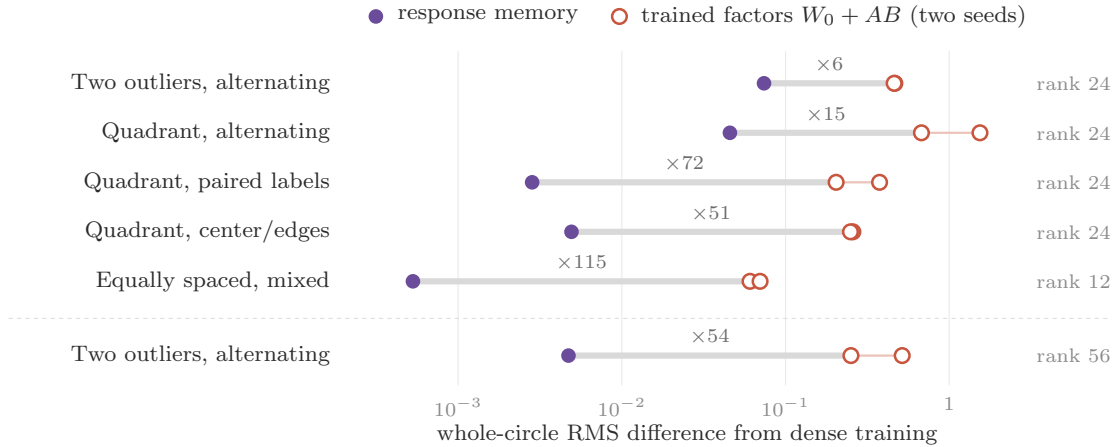


Figure 12: Same rank, different dynamics. Whole-circle RMS difference from dense training for response memory and for a directly trained correction $W_0 + AB$ of the middle matrix with the same rank (two factor seeds), two hidden tanh layers, width 2048. Rank 24 corresponds to $q = 3$, rank 12 to $q = 3$ on the four-representative antipodal quotient, and rank 56 to $q = 7$; labels give the smaller of the two ratios. Matched-loss endpoint comparisons against this specific factor baseline, not against every low-rank method; all ranks and the fitted functions are in Figure 13.

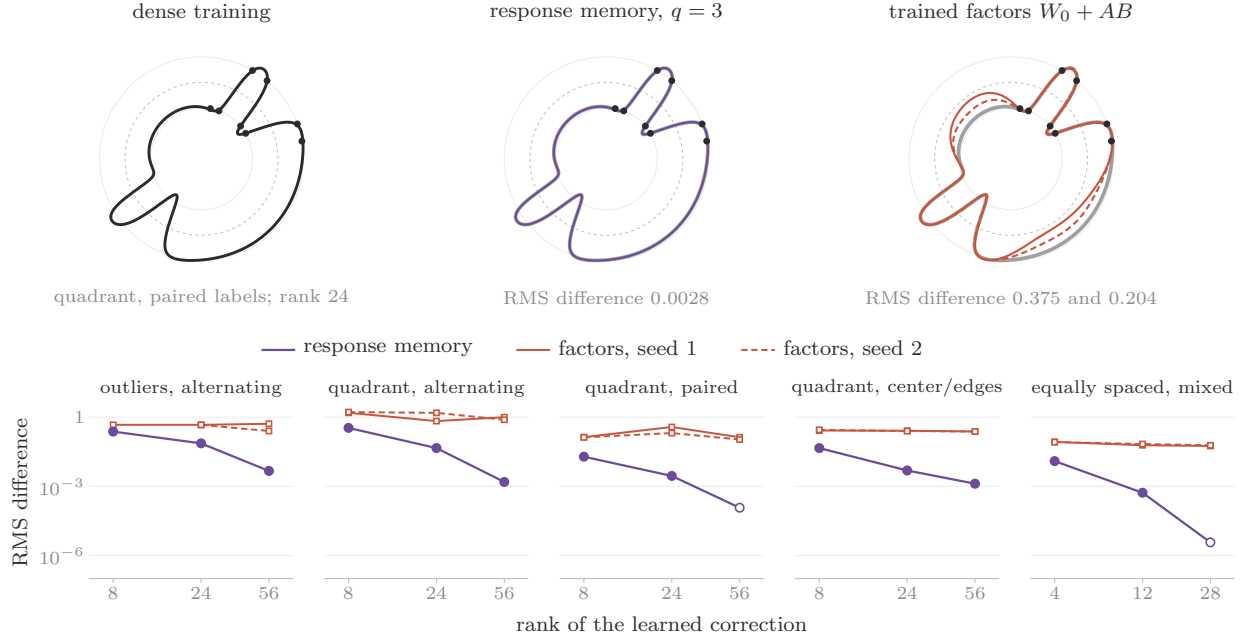
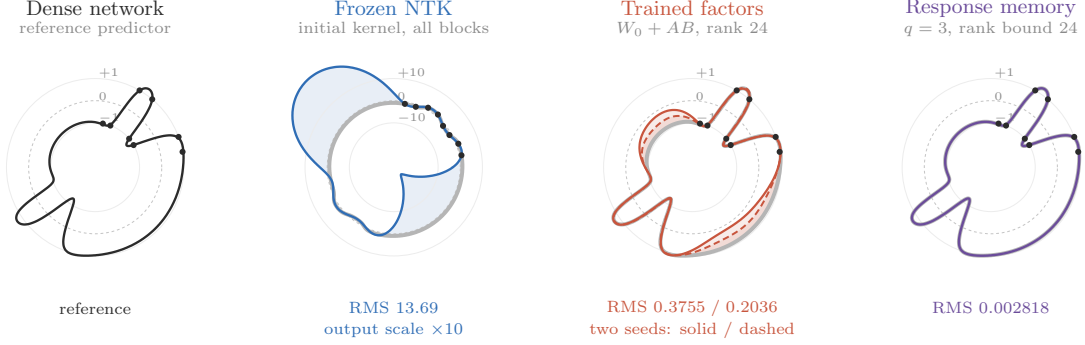
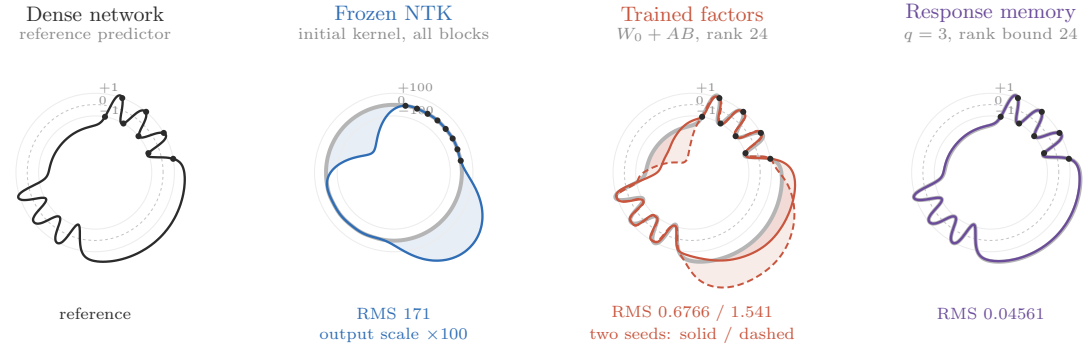


Figure 13: Same correction rank, different fitted functions. Two hidden tanh layers of width 2048. Top: paired-label task, memory $q = 3$ and both factor seeds at rank 24, with the dense reference in black. Bottom: all five tasks at rank bounds 8, 24, 56 (or 4, 12, 28 for the exact antipodal quotient). Every model is shown at its own MSE 0.001 endpoint; RMS uses 8192 angles. Factors use $W_2 = W_0 + AB$, $A(0) = 0$, independent $B_{ij}(0) \sim N(0, 1/\text{rank})$, and unit factor mobilities. Both outer layers have their canonical mobilities. Equal rank matches the factor-array budget, not total state or runtime. One capped comparison is omitted; hollow memory points have limited numerical precision.

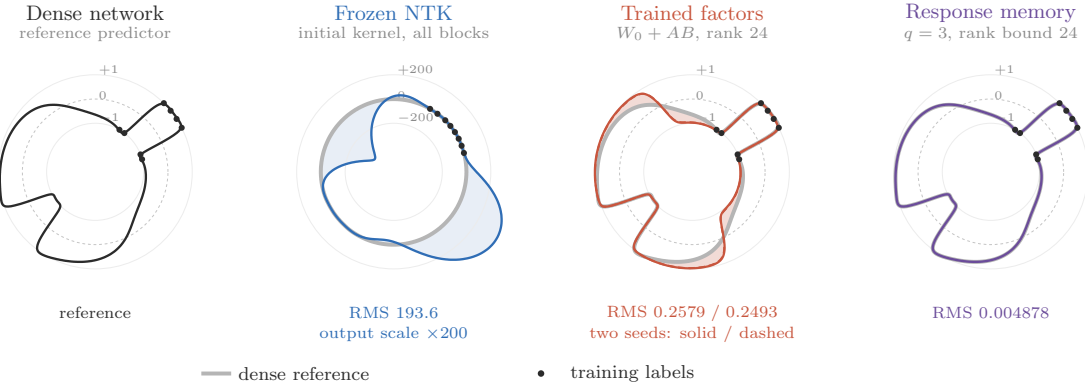
Quadrant, paired labels



Quadrant, alternating



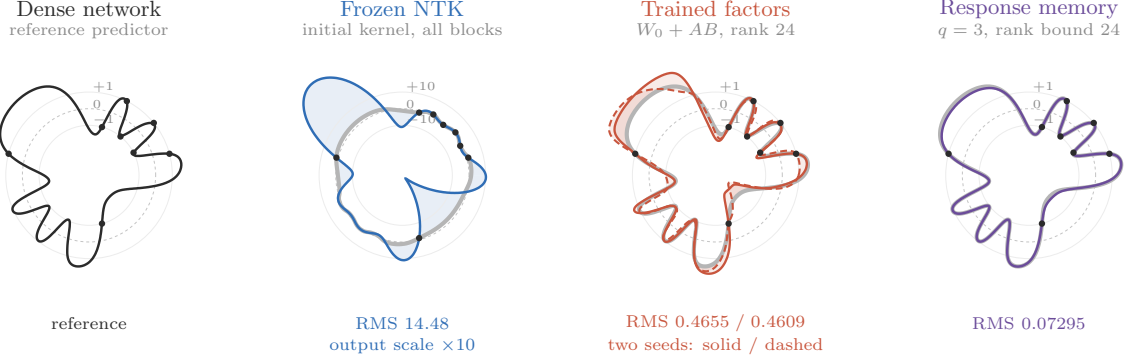
Quadrant, center/edges



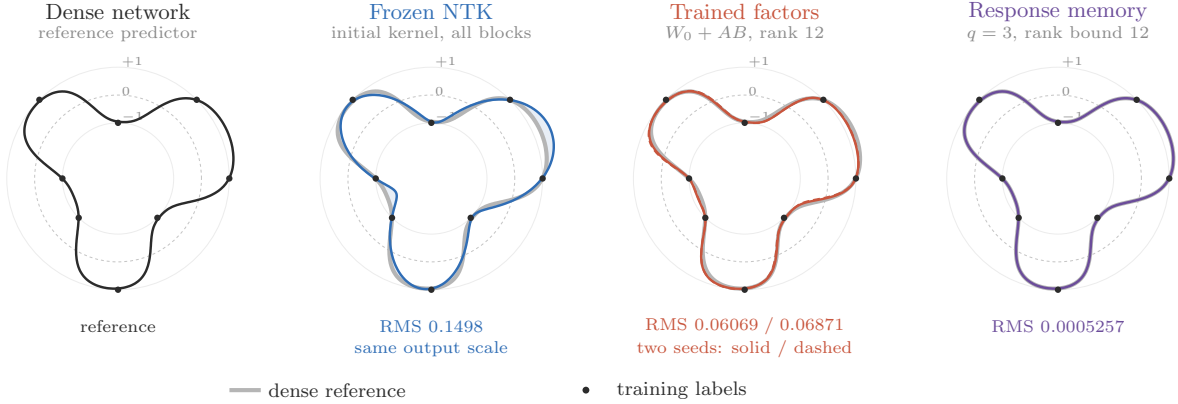
Same initialized network; each model at training MSE $\simeq 0.001$, at its own fitting time.
RMS differences from dense use 8192 circle queries. Signed ticks give output values; any wider NTK scale is marked explicitly.

Figure 14: Four training descriptions on five circle tasks, part one. Paired-label, alternating-label, and center/edges quadrant tasks. Columns show dense, frozen NTK, both directly trained factor seeds, and memory $q = 3$. The latter two have correction rank bound 24. Each model is at its own training-MSE 0.001 endpoint. The NTK output scale is wider by the marked factor; signed ticks and RMS retain original output units. Gray is the dense reference, black dots are training labels, and RMS uses 8192 circle queries. The next page continues with the remaining two tasks.

Two outliers, alternating



Equally spaced, mixed



Same initialized network; each model at training MSE $\simeq 0.001$, at its own fitting time.
RMS differences from dense use 8192 circle queries. Signed ticks give output values; any wider NTK scale is marked explicitly.

Figure 14: Four training descriptions on five circle tasks, continued. Two-outlier alternating labels and equally spaced mixed labels. The latter uses four antipodal representatives and correction rank 12; the former uses rank 24. All eight physical training inputs are displayed. NTK uses a ten-times wider output scale for the outlier task and the same scale as the other methods for the equally spaced task. Models, endpoints, seeds, colors and RMS definitions follow the preceding page.